

AUTOMATED PRODUCE SCANNER WITH REAL-TIME PRICING AND
PACKAGING

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CAPSTONE PROJECT ABSTRACT

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8. Abstract



8.1 Background of the Study

Manual vegetable weighing and pricing in retail environments remain susceptible to human error, inconsistent pricing, inaccurate measurements, and inefficient transaction processing. In many local supermarkets and public markets, the process of identifying vegetables, weighing products, computing prices, and labeling items is still manually performed, which can lead to delays and incorrect transactions. Existing automated systems commonly focus only on individual functionalities such as object detection, weight sensing, or barcode generation, and often lack integration into a complete real-time embedded solution capable of handling multiple retail operations simultaneously. Additionally, the absence of automated packaging and transaction monitoring may reduce operational efficiency and customer convenience.

This study aimed to address these limitations by developing an Automated Produce Scanner with Real-time Pricing and Packaging that integrates vegetable classification, weight measurement, price computation, barcode generation, and automated sealing into a single embedded platform using Raspberry Pi, Edge Impulse, and electromechanical components. The proposed system was designed to provide a practical and low-cost automated retail solution capable of improving operational efficiency, minimizing human intervention, reducing pricing errors, and enhancing customer experience in retail environments.



8.2 Summary

The study developed an automated vegetable classification, weighing, and labeling system capable of detecting red onion, garlic, and marble potato using a trained Edge Impulse model integrated with Raspberry Pi. The developed prototype combines computer vision, load cell sensing, barcode generation, database management, and automated packaging mechanisms into a unified embedded system designed for retail applications. The system performs real-time image detection to identify vegetables, measures weight using a load cell sensor, computes the corresponding price based on the stored unit price, generates and prints barcode labels, and performs automated sealing for packaging.

The prototype was evaluated using descriptive and quantitative methods based on classification accuracy, assorted detection accuracy, weight measurement accuracy and precision, price computation accuracy, barcode printing accuracy, sealing reliability, and overall system stability. A self-made questionnaire aligned with the objectives and major features of the prototype was utilized to assess the usability, functionality, and reliability of the system based on the respondents' experiences and observations. The questionnaire was validated by experts and subjected to reliability testing using Cronbach's Alpha, confirming excellent internal consistency of the evaluation instrument. The gathered data were analyzed using statistical methods such as mean and standard deviation to determine the acceptability and performance of the developed system.



8.3 Findings

The findings revealed that the system successfully achieved 100% accuracy in assorted vegetable detection and price computation, ensuring that mixed vegetables were correctly identified and prevented from being processed while maintaining accurate pricing operations. The overall vegetable classification accuracy reached approximately 80%, with performance varying depending on the vegetable type. The system demonstrated stronger recognition performance for red onion compared to garlic and marble potato, indicating that visually similar vegetables require further dataset enhancement and model optimization.

The weighing system produced generally accurate results within an acceptable tolerance range of ± 5 grams despite minor fluctuations caused by factors such as sensor sensitivity and load distribution. Barcode generation, printing, and scanning achieved complete functionality and reliability, successfully matching all printed codes with the corresponding database records. Initial inconsistencies observed in the automated sealing mechanism were improved through calibration and hardware adjustments, resulting in more stable and reliable sealing performance during later trials.

Furthermore, the evaluation results gathered from respondents indicated high acceptability of the developed prototype in terms of usability, functionality, and reliability. Respondents described the system as convenient, efficient, practical, and beneficial for local supermarket operations. The statistical evaluation also showed high



mean ratings and low standard deviation values, indicating strong agreement and consistency among the respondents regarding the performance and effectiveness of the developed system.

8.4 Conclusion

The study concludes that the developed Automated Produce Scanner with Real-time Pricing and Packaging is functional, reliable, and suitable for practical implementation in automated retail environments. The integration of computer vision, load cell technology, embedded systems, and electromechanical components successfully enabled accurate vegetable detection, real-time pricing, barcode labeling, and automated packaging within a single embedded platform. The system demonstrated the capability to automate several retail operations while reducing the need for manual intervention and minimizing transaction errors.

The developed prototype also demonstrated excellent performance in terms of assorted vegetable detection, price computation, barcode generation, and system stability. Although the classification accuracy for certain vegetables and the sealing mechanism still require further refinement and optimization, the overall performance of the system confirms the viability and effectiveness of the proposed innovation. The study further demonstrates that low-cost embedded technologies such as Raspberry Pi combined with machine learning models from Edge Impulse can be utilized to develop efficient,



accessible, and practical automated retail solutions for local supermarkets and similar retail environments.

8.5 Recommendation

Based on the findings and feedback gathered from respondents, the developed system is recommended for use in local supermarkets and retail environments because it improves convenience, operational efficiency, and transaction processing. Respondents emphasized that the system can make retail operations faster, easier, and more practical for daily use while reducing manual workload and improving customer experience.

Future studies may further improve the system by increasing the dataset size, particularly for visually similar vegetables such as marble potato and garlic, and by incorporating more diverse training conditions including different lighting environments, object placements, and background variations. Implementing controlled lighting conditions and improving image acquisition consistency are also recommended to enhance classification accuracy and overall detection performance.

Additionally, future researchers may improve the weighing system by applying filtering techniques to reduce fluctuations and increase measurement stability. Enhancements to the sealing mechanism, such as improved alignment and heat regulation, are also recommended to ensure more consistent packaging performance. The integration of cloud-based databases, multi-terminal support, and larger-scale retail deployment may further enhance the scalability and usability of the system.



Furthermore, the study supports Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure by promoting technological innovation and automation, and SDG 12: Responsible Consumption and Production by improving efficiency, minimizing operational errors, and supporting smarter retail practices through automated embedded systems.

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Chapter 1

INTRODUCTION

Background of the Study

The global retail industry continues to be transformed by the convergence of digital technology, shifting consumer behavior, and supply chain innovations. In particular, the application of automation and artificial intelligence in food retail is redefining traditional processes in inventory control, point-of-sale systems, and customer interaction. Globally, major supermarkets and retail chains have adopted smart checkout systems to streamline operations and reduce reliance on labor. The digitalization of in-store experiences, primarily through mobile applications and machine vision systems, enables retailers to collect real-time data on pricing patterns, consumer preferences, and inventory turnover. These tools reduce human error, improve pricing accuracy, and shorten wait times at checkout points.

Contemporary research on these global trends has identified both enabling and limiting factors in consumer adoption. One framework for understanding consumer interaction with automated checkout systems is the affordances flow funnel model. Studies show that consumers gravitate toward automated processes when they are intuitive, time-saving, and responsive to real-time data inputs (Naeem et al., 2024).

The Philippines faces unique challenges and opportunities in modernizing its food retail sector. In major cities such as Metro Manila and Cebu, supermarkets have begun introducing self-checkout machines and digital payment systems. The broader food retail



environment remains manual, particularly in areas where fresh produce is sold by weight or volume. This means that transactions are prone to pricing discrepancies and inefficiencies. A study on Filipino consumer behavior regarding food shopping patterns reveals that low-income households tend to frequent small, informal stores due to their proximity, affordability, and familiarity (Reimold et al., 2024). This reveals a gap that could be addressed through innovations that do not depend entirely on large-scale retail chains.

In Bacolod City, the relationship among urbanization, consumer density, and evolving retail formats provides a strong case for targeted automation. Bacolod has seen a rise in hybrid retail models combining formal supermarket chains with community-based produce vendors. Supermarkets in malls and commercial centers offer packaged goods, while produce sections are still labor-intensive. The goal is to implement self-service in the produce area. Automating the processes of scanning, pricing, and packaging produce within these venues could reduce customer wait times, improve price transparency, and enhance transaction accuracy.

There are also operational pain points experienced by retailers in Bacolod that automation could help alleviate. Manual weighing and pricing consume time and introduce inconsistencies. Packaging is often done reactively, resulting in missed opportunities for inventory tracking. The lack of real-time pricing adjustments also makes it difficult for vendors to respond to supply-and-demand fluctuations.



Introducing an automated scanner with real-time information capabilities could help mitigate impulsive overbuying and reduce food waste. With real-time scanning and packaging, Bacolod vendors could offer more precise discounts or promotions based on actual shelf life, rather than relying on arbitrary markdowns. Despite the availability of automated technologies, many existing systems focus only on individual components such as classification or weighing, and lack full integration into a single real-time embedded solution. Additionally, manual processes in vegetable retail remain prone to human error, inconsistent pricing, and inefficiency. Therefore, there is a need for a system that integrates image detection, weight measurement, price computation, and transaction recording into a unified and automated platform.

Objectives of the Study

To design, develop, and evaluate a functional prototype of an Automated Produce Scanner with Real-time Pricing and Packaging.

Specifically, this study had the following objectives, to:

1. Design and develop an automated vegetable classification, weighing, and labeling system:
 - a. Image detection;
 - b. Weight measurement;
 - c. Automated sealing mechanism;
 - d. Barcode sticker generation;

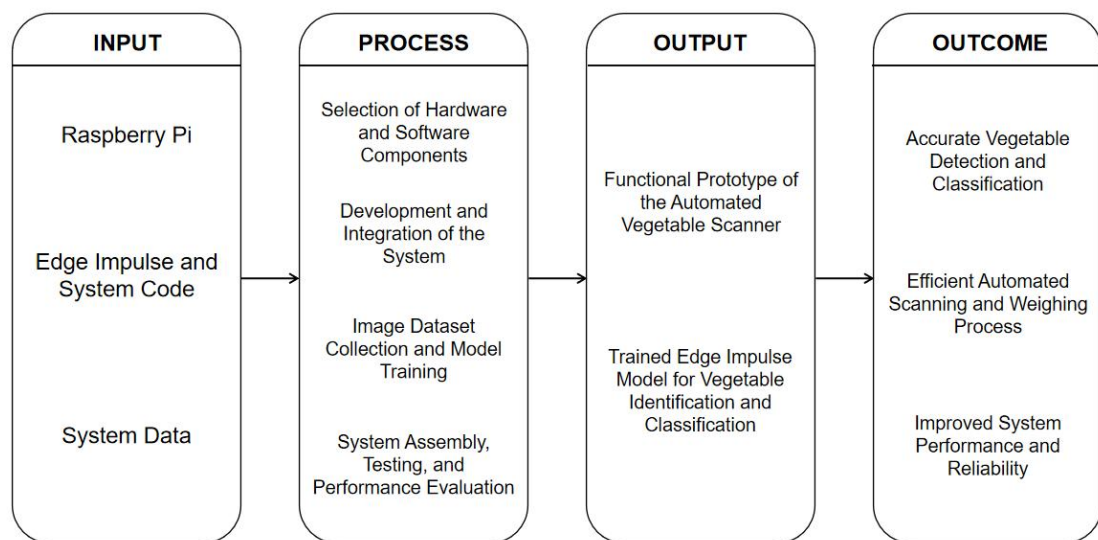


- e. User interface display;
- 2. Implement system functionalities for processing and data management, including:
 - a. Real-time price calculation;
 - b. Updatable price database;
- 3. Test and evaluate the performance, usability, functionality, and reliability of the integrated system in terms of:
 - a. Classification accuracy;
 - b. Assorted detection accuracy;
 - c. Weight measurement accuracy and precision;
 - d. Price computation accuracy;
 - e. Barcode printing accuracy;
 - f. Sealing reliability;
 - g. System stability;
- 4. Develop a user's manual.



Conceptual Framework

Figure 1. IPO Model



The Input-Process-Output-Outcome (IPOO) diagram presents the overall framework of the study for the development of the Automated Produce Scanner with Real-time Pricing and Packaging. The diagram illustrates the flow of the system from the required inputs, development processes, generated outputs, and expected outcomes of the study.



The input stage consists of the Raspberry Pi, Edge Impulse and system code, and system data. These components serve as the primary resources needed for the development and operation of the system. The Raspberry Pi functions as the main processing unit, while Edge Impulse and the developed system code provide the machine learning and automation capabilities required for vegetable detection and classification. System data includes the datasets, pricing information, and other necessary information utilized by the system.

The process stage includes the selection of hardware and software components, development and integration of the system, image dataset collection and model training, and system assembly, testing, and performance evaluation. During this stage, the researchers integrated the hardware and software components, trained the Edge Impulse model using collected vegetable datasets, and evaluated the performance and functionality of the developed prototype.

The output stage consists of the functional prototype of the Automated Vegetable Scanner and the trained Edge Impulse model for vegetable identification and classification. These outputs represent the completed system and its machine learning model capable of performing real-time detection and processing tasks.

Lastly, the outcome stage highlights the expected benefits of the study, including accurate vegetable detection and classification, efficient automated scanning and weighing processes, and improved system performance and reliability. These outcomes



demonstrate the practical significance and effectiveness of the developed system in supporting automated retail operations.

Scope and Limitations

This study develops and tests a self-service prototype that automates the purchase of three (3) specific vegetables: red onion, garlic, and marble potato; by using integrated machine learning (Edge Impulse) for image recognition. The system incorporates image detection to prevent the mixing of assorted vegetables; however, its performance is sensitive to lighting conditions. Under typical supermarket lighting, primarily cool white or daylight illumination (around 4000K–6500K), classification accuracy remains stable, but it is estimated to decrease by fifteen (15)% to twenty-five (25)% when lighting is insufficient or inconsistent (Chuquimarca, 2025). The prototype also features precise weighing and pricing, an error detection notice displays when multiple vegetables are detected, the ability to update produce prices, automated bag sealing, and barcode receipt generation from a local database using a Raspberry Pi, computer vision, and integrated electromechanical components.

This prototype has several significant limitations: it can only identify a fixed set of 3 local, small vegetables due to model training, and a wide variety of vegetables. The image detection can't detect that the vegetable is overripe or damaged. The weighing system can only accommodate at least a hundred (100) grams to three (3) kilograms of



weight produced, which may not be suitable for bulk customers. The user must choose one (1) kind of vegetable at a time. The system operates as a standalone unit with a local database and does not support multi-site or centralized system integration. Moreover, the packaging function is limited to specific heat-sealable plastic bags made of Low-Density Polyethylene (LDPE), and the overall system is designed for sequential operation.

Significance of the Study

The project will be beneficial to the following:

Consumers and Shoppers

This study will provide a faster, more accurate, and hygienic shopping experience by reducing wait times, ensuring correct weighing and pricing, and offering secure, automated packaging.

Retailers and Produce Vendors

This study offers a tool to increase operational efficiency, reduce labor costs and human error in pricing, improve inventory tracking, and handle customers more quickly, especially during peak hours.

Researchers and Future Developers

This study serves as a practical reference and prototype for developing similar automated systems, contributing to knowledge in computer vision, embedded systems, and affordable retail technology.



Local Communities and Public Markets

This study provides a scalable model for modernizing traditional retail settings, such as wet markets, by introducing digital tools that improve service quality and economic efficiency and support broader digitalization efforts.

Definition of Terms

The following terms are defined both conceptually and operationally to ensure clarity and consistency in understanding the study's development.

Automated Produce Scanner

The retail industry has undergone significant transformation through the integration of modern technologies designed to improve operational efficiency and customer satisfaction. One notable innovation is the implementation of self-checkout systems, which enable consumers to independently scan, bag, and pay for their purchases without requiring traditional cashier assistance. These systems improve convenience, reduce transaction time, and support efficient retail operations (Ali, 2024).



In this study, the automated produce scanner is the prototype developed by the researchers that identifies a vegetable, determines its weight, computes the price, and completes the packaging process as part of a self-service retail setup.

Load Cell with HX711

The Load Cell Amplifier is a compact breakout board featuring the HX711 integrated circuit, designed to simplify reading load cells for weight measurement (Mohite et al, 2025).

In this study, the load cell paired with the HX711 module serves as the weighing component, measuring the weight of each vegetable placed on the system and providing the data needed for real-time pricing.

Packaging System

Traditional packaging functions as a passive system designed to protect products from external hazards, highlighting the need for smart systems that enhance packaging's advantages. In contrast, active packaging aims to extend shelf life and maintain food quality, while intelligent packaging provides technologies for quality assessment and tracking, enabling seamless communication with consumers regarding the freshness, safety, and overall quality of the food (Palanisamy, 2024).

In this study, the packaging system refers to the automated sealing process that encloses the weighed and priced vegetable in a plastic bag after scanning.

Produce Identification



In recent years, computer vision technologies have drawn considerable attention in the field of precision agriculture. As a cornerstone of robotics and artificial intelligence, computer vision supports a range of tasks throughout the crop production cycle—from planting to harvesting—enabling them to be carried out automatically and efficiently (Lu & Young, 2020).

In this study, produce identification refers to how the system determines what type of vegetable is placed on the platform before computing its price.

Real-time Pricing

A central component of modern marketing involves developing a pricing strategy, which remains one of the four fundamental elements of the traditional marketing mix. Dynamic pricing offers a practical way to implement this strategy and is increasingly being adopted across various industries for two main reasons. First, gathering data on transactions and customer behavior has become feasible, cost-effective, and straightforward. Second, machine learning algorithms, which rely heavily on such data, are now widely accessible (Nowak & Pawlowska-Nowak, 2025).

In this study, real-time pricing refers to the system's automatic calculation of the total cost of a selected vegetable immediately after it is identified and weighed.

Edge Impulse

Machine learning on embedded and edge devices enables real-time inference directly on hardware with limited computational resources, eliminating the need for cloud processing (Warden & Situnayake, 2019).



In this study, Edge Impulse is used as a development platform for training, building, and deploying machine learning models. It serves as a tool for preparing datasets and enabling real-time object detection.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter presents and critically analyzes the existing literature and publications on the subject's functionality, with particular emphasis on technological features.



Machine Vision in Agricultural Product Identification

Recent advances in machine vision and deep learning have substantially influenced agricultural automation, particularly in the identification and classification of fruits and vegetables. The integration of artificial intelligence into agricultural contexts enables accurate and efficient product recognition, which is essential in streamlining retail, sorting, and packaging operations. Central to this progress are convolutional neural networks (CNNs), which have demonstrated superior performance in pattern recognition and feature extraction from image data. Pham et al. (2024) presented a deep learning-based IoT system that successfully recognizes fruits and vegetables by utilizing a CNN model deployed in an innovative embedded framework, offering high accuracy and low latency in recognition tasks. Similarly, Yan (2024) improved CNN-based models for vegetable classification, focusing on optimizing feature-extraction layers to handle intra-class variability and to enhance classification precision under normal lighting conditions.

Mushraf and Ahmed (2025) emphasized that CNNs have become the backbone of many fruit and vegetable recognition systems due to their ability to learn spatial feature hierarchies from input images. Their study highlights how layered convolution and pooling operations significantly contribute to robust feature representation, particularly in datasets with diverse visual patterns. Xiao et al. (2023) expanded on this by reviewing both traditional machine learning and digital image processing techniques used in harvesting robots, indicating that while conventional approaches remain valuable, deep learning models like CNNs and YOLO architectures are increasingly preferred due to



their scalability and adaptability. Oliullah et al. (2025) introduced FruVeg_MultiNet, a hybrid IoT system that incorporates CNN-based recognition with a user interface and assistive hardware for the visually impaired. This integration emphasizes the potential of machine vision beyond automation, extending into inclusive technology and accessibility. The presence of a web-based interface also reinforces the role of responsive visual systems in real-time environments.

The success of these intelligent systems relies heavily on the foundational understanding of object detection frameworks. Hou et al. (2023) provided a comprehensive overview of vision-based recognition and localization methods for harvesting robots, noting that successful implementations must combine high detection accuracy with real-time processing. They highlight the persistent challenge of balancing model complexity with computational efficiency, especially when deployed in embedded hardware with limited processing power. Zargham et al. (2023) addressed this issue in a retail context by proposing an IoT-based intelligent scale system for small-scale fruit and vegetable vendors. Their solution uses embedded machine vision modules to classify produce, enabling real-time pricing and packaging without manual intervention. Their findings illustrate the practicality of transferring machine vision from large-scale automation to more minor, decentralized commercial operations.

In similar studies, Kang et al. (2024) explored how deep learning techniques enhance object localization and detection, proposing pipeline improvements that reduce false positives while maintaining real-time throughput. Their emphasis on dataset



variability supports the argument that model training requires well-annotated, high-resolution image data. This is echoed in the work of Pandey et al. (2023), who performed a comparative analysis between YOLO v7 and YOLO v4 for fruit detection. Their results favored YOLO v7 due to improved bounding box regression and object tracking under variable lighting and occlusion conditions. The need for lightweight and sustainable models is also emphasized by Gaurav et al. (2025), who developed a CNN framework tailored for innovative food systems. Their model achieves high classification accuracy with low computational overhead, making it ideal for embedded deployment in retail scanners.

In pursuit of computational efficiency, Khoiruddin and Tena (2024) adopted MobileNetV2, a lightweight CNN architecture, to classify fruits and vegetables. Their results indicate that even with limited hardware resources, MobileNetV2 offers sufficient precision for practical applications. Srivats et al. (2024) expanded the discussion by combining clustering algorithms with vision models to improve the classification of produce based on visual similarity. This hybrid method enhances the system's ability to distinguish subtle differences between similar-looking items, which is particularly useful in sorting tasks. Manalo et al. (2025) applied a machine learning framework to automatically grade and sort eggplants, demonstrating that tailored feature engineering can compensate for challenges posed by dataset size and visual consistency. Their work further bridges the gap between computer vision and physical actuation, demonstrating



how classification results can trigger downstream mechanical responses, such as sorting or packaging.

In another related domain, Ulloa et al. (2022) applied CNNs to detect and characterize vegetable types during robotic fertilization in strip cropping systems. Their results highlighted the dual benefit of visual detection for both classification and agronomic decision-making, revealing broader implications for precision agriculture. Raza et al. (2025) introduced an intelligent image gallery system powered by deep learning for automated fruit and vegetable classification. Their focus on gallery-based approaches provides a novel method for real-time matching and validation of classified produce. This contributes to traceability and consistency in commercial retail systems. The foundational infrastructure supporting these systems often involves IoT technologies. As discussed by Kuppam and Purushotham (2021), recent IoT advancements have enabled greater connectivity, sensor integration, and remote data access across agricultural applications, reinforcing the feasibility of deploying such systems in live retail or farm environments.

Amini and Rahmani (2023) discussed the role of agricultural databases and their assessment through machine learning. They noted that poorly structured or biased datasets can significantly impair model generalizability and performance. Similarly, Meshram and Patil (2022) introduced FruitNet, a curated Indian fruits dataset intended for machine learning applications, emphasizing the importance of quality labeling and image diversity. In a broader scope, Benos et al. (2021) provided a comprehensive review



of machine learning applications in agriculture, establishing that vision-based tasks are among the most impactful due to their direct applicability in automation and quality control. They argue that the convergence of hardware advances, dataset availability, and algorithmic innovations is propelling agricultural AI into more practical, field-deployable solutions. Complementing this, Meshram et al. (2021) surveyed the state of machine learning in agriculture, confirming that object recognition for fruits and vegetables is a mature domain ripe for commercial scaling.

AI-Driven Produce Recognition and Automated Pricing Systems

Recent developments in artificial intelligence and embedded systems have enabled fully automated retail billing processes, particularly for fruits and vegetables that lack barcodes. Jayashree et al. (2021) developed an intelligent vegetable processing and pricing system that uses Convolutional Neural Networks (CNN) for produce recognition and cloud-based APIs for dynamic price updating. This allowed the system to automatically adjust prices based on current market conditions, reducing manual database entry and eliminating inconsistencies caused by outdated pricing. Similarly, Azizan and Md Nor (2025) introduced FoodSnap, a mobile application that applies image recognition to instantly retrieve food prices from a MySQL database, improving price transparency in cafeteria environments where price tags are often missing or inaccurate. Their work shows that image-based billing can empower customers by allowing them to verify the price of an item before purchase, which is especially beneficial for budget-conscious



users such as students. Kumar et al. (2024) further enhanced retail efficiency with an IoT-based smart shopping cart that enables customers to scan product barcodes directly on the cart, wirelessly transmitting billing data to a central server for fast payment processing. Their system demonstrates that decentralizing the checkout process can significantly reduce congestion at payment counters and improve the overall shopping experience. In another study, Shriyan et al. (2025) demonstrated that deep learning models, such as Faster R-CNN, outperform faster architectures, such as SSD, in grocery billing accuracy, making them more suitable in environments where even minor pricing errors can affect business credibility and customer trust.

Artificial intelligence-driven produce identification has also been applied to small-scale retail environments, extending these innovations beyond large supermarket chains. Zargham et al. (2023) developed an IoT-integrated weighing scale combining YOLO-based classification and load-cell-based weight measurement to automate produce pricing in small vegetable shops, significantly reducing reliance on manual weighing and cashier input. Their system serves as evidence that advanced image recognition technology can be applied even in informal market settings. Likewise, Vennila et al. (2021) implemented a Raspberry Pi-based billing system using Support Vector Machines (SVM) for fruit recognition paired with load-cell-based weighing, proving the feasibility of affordable computer-vision-assisted checkout systems using low-cost embedded platforms. Meanwhile, Kiplagat (2024) introduced a hybrid system that combines CNN classification with Bayesian inference to improve decision-making under uncertain visual



conditions during grocery checkout, reinforcing the potential of AI-assisted automated billing when environmental factors such as lighting or item similarity create ambiguity. Complementing these contributions, the systematic review by Rojas Santelices et al. (2025) confirmed that deep learning has become the dominant approach in fruit recognition and inspection, supporting the expansion of artificial-vision technologies from agricultural fields and industrial processing lines into consumer retail environments.

Electromechanical and Mechatronic Automation in Food Packaging Systems

Automation has also transformed food packaging processes, improving hygiene, consistency, and operational output across both industrial and small-scale settings. Tianfei (2020) emphasized the importance of mechatronic control in modern food packaging machinery, explaining that automated sealing and cutting systems ensure sanitary and uniform packaging compared to traditional manual methods, which are often inconsistent and labor-intensive. These systems integrate mechanical hardware with electronic control modules, enabling packaging tasks to be performed more quickly, accurately, and repeatably. Automated packaging is also closely linked to food safety, as uniform sealing reduces contamination risks and helps maintain product freshness during storage and transport.

In the same vein, Hambir et al. (2019) designed a low-cost automatic weighing and packaging system that uses a hopper, a load cell, and a servo-controlled dispensing mechanism to seal products using a heating element. Their findings demonstrated that



automating these processes significantly reduces operator fatigue, increases production throughput, and minimizes human error, all without requiring expensive industrial-grade equipment that would normally be beyond the reach of small retailers. To further improve seal consistency and reduce worker strain, Shaharuddin et al. (2025) developed a semi-automatic sealing system controlled by an Arduino microcontroller and Infrared (IR) sensors, enabling reliable and uniform seal application in Small and Medium Enterprises (SMEs). The system detects the presence of a packaging bag and automatically applies precise sealing pressure via a pneumatic actuator, ensuring that each seal meets hygiene and durability standards regardless of the operator's skill level. Collectively, these studies demonstrate that electromechanical automation not only enhances operational efficiency but also improves product safety, extends shelf life, and enhances the professional presentation of packaged goods. These findings provide strong support for integrating automated sealing mechanisms into modern retail systems, particularly in developing markets where packaging processes are still commonly performed manually.

Image Processing–Based Morphological Classification

In a local study conducted by Arboleda et al. (2024), the researchers developed a computer vision system to classify eggplant varieties using image processing techniques. The system extracted morphological features such as area, perimeter, and roundness from digital images, which were then processed using K-Nearest Neighbors (KNN) and Fuzzy Logic for variety classification. Their findings showed that geometric feature extraction is



an effective method for identifying agricultural produce, supporting the feasibility of camera-based identification in the Philippine setting. However, while their work focused only on eggplant varieties, the proposed study aims to classify multiple types of vegetables using Convolutional Neural Networks (CNN), which are more suitable for handling diverse and complex produce categories.

Technology Acceptance of Self-Service Kiosk Systems

In a local study conducted by Peña and Aquino (2023), the researchers examined how Filipino customers perceive and accept the use of self-ordering kiosks in fast-food restaurants. Using the Technology Acceptance Model (TAM), they surveyed 300 respondents to evaluate perceived usefulness and ease of use compared to traditional counter service. The results showed that most customers preferred kiosks because they reduced waiting time and made transactions more convenient, although some still expressed concern about possible technical errors. The study concluded that self-service technology is generally well-accepted, provided that the system remains stable and reliable during operation.

Vision-Based Self-Checkout and Automated Pricing Systems

In a recent study by Zhang et al. (2024), the researchers developed a fruit and vegetable self-billing system that uses image recognition to automate pricing for non-barcode supermarket items. Their system runs on a Raspberry Pi and applies a



Convolutional Block Attention Module (CBAM), allowing the model to focus on key visual features of the produce while reducing background interference. The researchers reported a high accuracy rate of 93% in identifying both freshness and type, enabling real-time and human-free billing transactions. This study demonstrates that embedded AI systems can reliably support automated pricing of produce in retail environments.

Fruit and Vegetable Identification Using Machine Learning

Although not among the listed local literature sources, the work by Femling et al. (2018) complements the study by Zhang et al. (2024) by demonstrating how machine learning and embedded systems can be used to automate produce identification in retail environments. Their system employed a Raspberry Pi, a camera, and a load cell, and tested multiple convolutional neural network architectures to classify fruits and vegetables. The study emphasizes improving the usability and speed of automated labeling systems while retaining accuracy in identification—findings that parallel the goal of the GRFO-MCNN model described by Zhang et al. Together, these works illustrate ongoing advancements in computer vision for retail applications, particularly how deep learning models can replace manual produce labeling and billing processes.

Seal Integrity in Heat-Sealed Flexible Food Packages

Ilhan et al. (2021) provide an extensive review of heat-sealing mechanisms and factors affecting seal integrity in flexible food packaging. Their study explains that the



seal area of food packages is the most critical region for preventing moisture and gas ingress, which directly affects food quality and shelf life. The review outlines various heat-sealing methods, including seal bars, hot wire, induction, hot air, pneumatic, and ultrasonic sealing, and highlights how process parameters such as temperature, pressure, and dwell time govern melting and diffusion at the seal interface. Importantly, the authors identify that material properties (e.g., polymer type, molecular weight, crystallinity), contaminants (liquid and solid particles), and structural design factors (like seal bar geometry and film tension) can create weak spots or microchannels that lead to leaks. They emphasize that even microscopic imperfections significantly reduce effectiveness, making it essential to understand these variables to improve packaging performance and reduce food waste. This review not only categorizes these factors but also discusses their interrelationships and how they influence seal formation and failure, providing a roadmap for future research in flexible food packaging technology.

REVIEW OF RELATED STUDIES

Load Cell Calibration and Real-Time Weight-Based Pricing Systems

The convergence of Internet of Things (IoT), embedded systems, and machine learning has significantly transformed traditional retail and agricultural operations. One of the most vital applications of this convergence lies in the deployment of real-time weight-based pricing systems using load cells and HX711 modules, along with the automation of receipt generation and packaging. Munna and Mallick (2024) described



how IoT-enabled intelligent weight machines connected to centralized pricing databases can streamline the pricing mechanism, enabling more consistent, real-time evaluation of product costs. Their system employed load cells paired with HX711 modules for analog-to-digital conversion, enabling accurate and rapid weight capture. This configuration supports immediate pricing decisions and enhances operational transparency. The emphasis on precise calibration techniques was central to their system, acknowledging that even minor sensor deviations can compromise pricing accuracy.

Expanding on these principles, Manoj et al. (2025) used YOLOv10 object detection on a Raspberry Pi to develop an innovative billing system capable of visually identifying products and weighing them in real time. The integration of load cell data with computer vision improved both item recognition and price tagging. Their design also implemented a local database for price look-up, which mirrors retail environments that demand offline functionality in low-connectivity areas. Similarly, Sawant et al. (2025) introduced an innovative retail shelf system with embedded sensors and weighing mechanisms, streamlining inventory monitoring and replenishment. Their system included real-time alerts for stock depletion, which not only supported retail efficiency but also aligned with sustainability goals by reducing overstocking and wastage. They also incorporated an HX711 amplifier module to enhance the sensitivity of their load cells, allowing for more granular data tracking.

In the field of secure and automated commercial technologies, Kowshik et al. (2025) presented an alternative locker system featuring weight detection and RFID



integration, showing how load cell-based mechanisms can extend beyond pricing and inventory into secure access control. While not directly tied to retail sales, their model emphasized the scalability and reliability of load cell platforms, particularly when integrated with IoT protocols. Their use of calibration routines for improved measurement precision directly contributes to the design integrity of retail weight-based systems. Meanwhile, Kumar et al. (2024) proposed an IoT-based smart shopping cart with an integrated payment gateway. Their implementation included load cells managed by HX711 converters, supporting a self-service retail experience. Their model promoted shorter checkout times, fraud prevention, and streamlined customer flow, underscoring the increasing relevance of integrated systems in real-world retail contexts.

Automation in Retail Packaging and Receipt Generation

The receipt generation component of retail automation is equally critical, especially in regulatory and fiscal applications. Kedia et al. (2024) developed an Arduino-based thermal mini-printer, demonstrating a low-cost, programmable solution for instant receipt generation. When integrated with pricing systems, this technology can offer customers a complete transactional experience without human intervention. Özhan (2025) provided historical and technical insights into barcode technologies, emphasizing their role in identification, traceability, and system interoperability. Barcodes function as a bridge between physical goods and digital transaction records, and their role is magnified when receipts include encoded data for customer tracking and inventory audits.



Merino et al. (2025) examined coercion-resistant registration systems within e-voting. Yet, their findings on digital verifiability and system usability have implications for fiscal receipt verification, as similar standards apply to retail transparency and auditability.

Bagal et al. (2025) introduced a QR-based bus pass system where user transactions and travel data were logged and accessed in real time. The conceptual model behind this is transferable to retail, where transaction histories can be stored using QR or barcode receipt formats. Talikovs and Litvinenko (2021) discussed fiscalization instruments relevant to digital economies, proposing that receipt generation must not only satisfy customer needs but also conform to taxation and audit regulations. Their emphasis on standardized formats and backend validation mechanisms can inform the design of automated pricing and receipt systems, particularly in regulated retail spaces. Gowthami et al. (2025) further explored secure innovative shopping applications, advocating for IoT-based systems that support both customer-facing and administrative interfaces. Their system included user authentication layers and secure data handling, key factors in receipt issuance and transaction protection.

Security in retail automation requires robust protocols, particularly with the increase in networked IoT devices. DR et al. (2024) proposed the I-Paver architecture, a QR-based system within OPC UA standards, to enhance IoT retail security. Their model supports secure communications between device nodes, protecting price data, weight readings, and receipt transactions. This is particularly relevant as pricing systems become increasingly automated and interconnected. Nagaty (2023) provided a broader view of



commercial and industrial IoT applications, including AI-powered pricing and packaging systems. His review emphasized the criticality of security, latency control, and power efficiency, all of which influence how load cells, printers, and administrative dashboards operate in real-time environments.

Authentication is another area of concern. Yalli et al. (2023) described a three-tier authentication system for IoT devices, ensuring secure access to sensitive modules such as pricing databases and transaction histories. Their system prevents unauthorized manipulation of product data, which is vital for ensuring accurate receipts and pricing integrity. Kumar and Nakkeeran (2023) analyzed denial-of-service vulnerabilities in IoT infrastructures, identifying attack vectors that could disrupt weight readings, price computation, or receipt printing. They emphasized the importance of feature selection and anomaly detection to secure embedded systems against intrusion or manipulation.

Although image recognition is not the primary function in weight-based pricing, its synergy with weighing modules enhances overall automation. Jin et al. (2025) used image-classification neural networks to detect weeds in vegetables, demonstrating the potential of deep learning to distinguish visual agricultural patterns. Waseem et al. (2025) developed a rapid prediction model for tomato maturity using image recognition, contributing to sorting and grading systems that can be linked with real-time pricing. Patel et al. (2025) designed a vegetable detection model optimized for market conditions, highlighting how visual classification and weighing can co-function to produce more precise pricing decisions. These models complement the work of Habib et al. (2021), who



reviewed machine vision methods for fruit and vegetable disease recognition, providing insight into the fine-grained classification techniques that support downstream pricing or packaging processes.

Mortadza et al. (2024) employed a Single Shot Detector on a Raspberry Pi to recognize vegetables in real-time. Their study aligns directly with applications in point-of-sale systems where item detection precedes weighing and pricing. This compact and energy-efficient approach validates the integration of visual and load-based inputs within embedded systems, particularly in decentralized or mobile retail settings.

Automatic Fruit Recognition

In a study conducted by Abir et al. (2022), the researchers focused on automating fruit recognition using deep learning to support retail and industrial applications. They observed that traditional image-processing methods struggle with variations in lighting, shape, and occlusion, which often leads to misclassification of fruits. To address this issue, the proponents developed a convolutional neural network–based framework that automatically extracts features from fruit images and classifies them into predefined categories. The system was trained and tested on a large dataset of fruit images, achieving high accuracy while keeping computational complexity relatively low. Their findings indicate that deep learning offers a reliable foundation for building automated fruit-scanning systems that can operate in real-world environments.



MobileNetV2 Fruit and Vegetable Recognition

In a study conducted by Almadhor et al. (2025), the researchers addressed the need for lightweight yet accurate models for fruit and vegetable recognition in real-time applications. They noted that many existing deep-learning models require powerful hardware, making them unsuitable for embedded devices commonly used in retail kiosks and smart carts. To mitigate this limitation, the proponents implemented a MobileNetV2-based classifier that recognizes multiple fruit and vegetable classes from images captured under varying conditions. The model obtained high classification accuracy while maintaining low inference time and reduced memory usage, demonstrating its suitability for deployment on mobile and edge devices. This work shows that efficient architectures such as MobileNetV2 can enable portable produce scanners that operate quickly without expensive computing resources.

Fruit and Vegetable Identification for Retail

In a study by Hernández-Jaramillo et al. (2018), the researchers explored how machine learning can simplify the identification of fruits and vegetables at retail checkouts. They observed that cashiers and customers often struggle with manual look-up of produce codes, which slows down transactions and increases the risk of pricing mistakes. To solve this, the proponents proposed an image-based identification system that captures an image of the produce item and applies a trained classifier to automatically determine its class. The recognized class is then linked to the store database



so that the correct price per unit or per kilogram can be retrieved without manual entry.

The results suggest that image-based identification can streamline POS operations and serve as the basis for automated produce scanners.

Intelligent Vegetable Processing and Pricing

In a study conducted by Nguyen and Tran (2024), the researchers addressed the challenge of integrating classification, weight estimation, and dynamic pricing for vegetables in a single automated system. They noted that many small retailers still rely on separate devices and manual calculations, which slows down service and limits the ability to update prices in real time. To overcome this issue, the proponents proposed an intelligent vegetable processing and pricing framework that uses computer vision to identify the vegetable, sensors to estimate weight, and cloud-based APIs to retrieve and update prices dynamically. The prototype demonstrated that the system can automatically compute and display the total cost while reflecting current market prices. This study directly supports the concept of an automated produce scanner with real-time pricing by showing that technical integration of recognition, weighing, and pricing is achievable.

General Automatic Fruit Recognition Framework

In a study conducted by Abir et al. (2022), the researchers sought to design a simple yet efficient automatic fruit recognition framework suitable for real-world deployment. They observed that many existing models are either too complex for



low-cost hardware or too sensitive to changes in illumination and background. To resolve this, the proponents developed a convolutional neural network that balances accuracy and computational cost by optimizing network depth and feature extraction. Experiments on multiple fruit datasets showed strong recognition performance while maintaining reasonable processing time. The results imply that carefully designed CNN architectures can be embedded into compact scanning devices for fast and reliable produce identification.

Mango Sweetness Assessment

In a study conducted by Dar et al. (2023), the researchers examined how computer vision can be used to estimate the sweetness level of *Mangifera indica* (Guimaras mango) without destructive testing. They pointed out that conventional laboratory methods for measuring sweetness are time-consuming and impractical for routine grading in supply chains. To address this concern, the proponents designed a system that segments mango images, extracts color features, and processes them using K-means clustering and a convolutional neural network to classify fruits into sweetness categories. Experimental results showed that the method achieved an accuracy of approximately 83.33% in distinguishing among the Super Sweet, Sweet, and Sour classes. This demonstrates the potential of image-based techniques for quality grading of



Philippine produce, which can later be integrated with automated pricing and labeling modules.

Embedded Gateway for Quality Classification

In a study conducted by Bansod et al. (2022), the researchers investigated an embedded smart gateway framework for real-time classification of fruit and vegetable quality. They observed that many small and medium enterprises lack affordable systems that can automatically assess produce quality during sorting and packaging. To overcome this challenge, the proponents developed a low-cost embedded platform that uses image sensors and machine-learning models to classify items into different quality grades as they pass through the gateway. The system processes images on-device and transmits classification results to external systems for logging or actuation, enabling real-time decisions about acceptance, rejection, or re-packaging. Their work highlights how embedded vision systems can be applied in small-scale operations, aligning with the needs of local markets that may adopt automated scanning, pricing, and packaging solutions.

Postharvest Processing Using Computer Vision

In a study conducted by Bhelkar et al. (2022), the researchers explored the application of computer vision technology in postharvest processing of fruits and vegetables, focusing on shape recognition algorithms. They pointed out that manual



inspection during sorting and grading is labor-intensive and can lead to inconsistent quality standards. To address this concern, the proponents implemented image-processing techniques that analyze the shape and size of produce items to enable automated sorting. The system successfully differentiated products based on geometric features and demonstrated potential for integration into postharvest handling lines. This work demonstrates how vision-based inspection can be used prior to pricing and packaging, ensuring that only properly graded items proceed to automated labeling and billing.

Real-time Guava Grading System

In a study conducted by Li et al. (2023), the researchers developed AVIG, a real-time visual inspection system for grading guava using computer vision and XGBoost. They observed that traditional grading approaches rely heavily on human inspectors, which can result in inconsistent quality and slow throughput. To mitigate these issues, the proponents designed a dual-view imaging system that captures guava fruits from different angles and extracts features related to size, color, and surface defects, which are then fed into an XGBoost model for grade prediction. The system achieved high grading accuracy and operated fast enough for use on sorting lines. Their findings demonstrate that real-time, camera-based grading is practical and can be combined with downstream modules for automated pricing and packaging.



Summary of Insights

The reviewed literature supports the objectives of developing an automated system for fruit and vegetable identification, real-time weight-based pricing, and transaction processing by demonstrating the effectiveness of machine vision, deep learning, and IoT technologies in retail and agricultural applications. Studies show that CNN- and YOLO-based models can accurately classify produce in real time, while lightweight architectures such as MobileNetV2 enable deployment on embedded platforms. The integration of load cells with HX711 modules has been shown to provide precise weight measurements essential for accurate pricing, with proper calibration identified as a critical factor in system reliability. Additionally, research on automated receipt generation using thermal printers, barcodes, and QR codes highlights their role in enhancing transaction transparency, traceability, and security. However, the literature also reveals that most existing systems address these components individually and lack full integration under real-world conditions, particularly in localized contexts. These findings directly support the objective of designing a fully integrated, cost-effective, and locally adaptable system that combines produce recognition, weight-based pricing, secure transaction handling, and receipt generation for practical deployment in the Philippine setting.

Across foreign studies, researchers consistently demonstrate that computer vision and deep learning can reliably recognize fruits and vegetables in images and enable automatic identification in retail or industrial environments. Lightweight architectures,



such as MobileNetV2 and other optimized CNN models, can run on embedded or mobile devices, which is important for real-time production scanners with limited hardware resources. Several works extend recognition by integrating weighing units and pricing modules, demonstrating that systems can automatically compute charges once the item type and weight are known, reducing manual code entry and pricing errors at the point of sale. Local and regional studies on mango sweetness assessment, guava grading, postharvest inspection, and embedded smart gateways demonstrate that similar vision-based methods and low-cost hardware are feasible for real-time quality grading and decision-making, especially for small and medium enterprises.



Chapter 3

METHODOLOGY

Research Design

This study utilizes a developmental research design, specifically following the Design and Developmental Research (DDR) framework, which is appropriate for studies that focus on the systematic creation, integration, and evaluation of a functional technological prototype. The primary objective of the research is to design, develop, and validate an Automated Produce Scanner with Real-time Pricing and Packaging, implemented using embedded systems, machine learning-based image detection, and electromechanical components.

The research design is structured into sequential and iterative phases, namely system development, implementation, evaluation, and validation, to ensure that each subsystem functions correctly before full system integration (Jaya, 2021). The developmental phase focuses on hardware selection and configuration, dataset preparation, model training, and software implementation. Core components used include a Raspberry Pi 4 Model B (4GB RAM) as the central processing unit, a Raspberry Pi Camera Module V2 (IMX219)

Image data are collected using the Raspberry Pi Camera Module mounted at a fixed height above a blue background platform under stable daylight illumination. The dataset consists of multiple vegetable classes, specifically red onion, garlic, and marble potato, along with an empty class representing the absence of produce. It also includes an



error detection that occurs when two (2) or more products are present on the weighing platform. Images are uploaded, labeled, and processed in Edge Impulse Studio, where a transfer-learning-based image detection model is trained and evaluated. The trained model is exported as a Linux (AArch64) deployment package (.eim file) and deployed for offline inference on Raspberry Pi using Edge Impulse with INT8 quantization, ensuring efficient and reliable execution on embedded hardware without cloud dependency.

The implementation phase integrates image detection with weight sensing. Upon detecting the presence of a red onion, the system retrieves weight data from the load cell, computes the total price based on a predefined price per kilogram, and displays the results on the LCD in real time. A tare button is incorporated to allow accurate zeroing of the weighing system before measurement. The software logic is implemented in Python, using OpenCV for image handling and system-level commands for camera control, ensuring compatibility with Debian 13 (Trixie) and the modern libcamera framework.

The evaluation and validation phase involves systematic testing of individual modules and integrated system behavior. Quantitative performance metrics such as classification or detection accuracy, weight measurement error, response time, and system stability are collected through repeated trials under controlled conditions. Functional validation also includes verifying correct LCD output, proper handling of error states (e.g., no produce detected and multiple vegetables detected), and accurate price computation based on measured weight. In addition to technical testing, a self-developed questionnaire is administered to selected users to gather structured feedback



on system usability, the clarity of displayed information, and the overall user experience.

The combined use of quantitative performance data and questionnaire-based evaluation supports a comprehensive assessment of the system's reliability, usability, and operational consistency.

Design Criteria

The design of the system was guided by the following criteria:

1. **Image Detection** - The system processes image frames using the trained Edge Impulse model, which outputs detected object labels along with corresponding confidence scores. A predefined confidence threshold is used to determine valid detections. If a single vegetable type exceeds the threshold, it is considered valid. However, if two or more different vegetable classes exceed the threshold within the same frame, the system identifies this as an “assorted vegetable” condition and triggers an error response, preventing further processing.
2. **Weight Sensing** - The weighing mechanism must provide precise measurements of produce weight up to a maximum of three (3) kilograms, with an acceptable tolerance of ± 5 grams (Troemmer, 2024). Accurate weight measurement is essential for accurate price calculation and maintaining trust in the automated checkout system. This criterion ensures financial accuracy and compliance with standard retail practices.



3. Sealing Mechanism - The packaging module must securely seal the produce in heat-sealable low-density polyethylene (LDPE) bags using an Impulse Sealer. The sealing process should ensure airtight packaging to prevent spillage, contamination, or tampering, and must operate consistently under the expected output of sequential produce processing.
4. Barcode Printing - The system shall integrate a thermal printer dedicated to printing a scannable barcode for each transaction. This barcode serves as the primary output for the cashier, encoding a unique transaction ID that links directly to the full purchase details, including item, weight, and price, stored securely in the system's database. The system shall integrate a thermal printer to generate a scannable barcode for each transaction. The barcode follows the European Article Number (EAN-13) standard, a 13-digit numeric format used in automatic identification systems to ensure reliable and efficient data capture (ISO/IEC 15420, 2009). The printer produces a clean, high-contrast barcode in a reliable format such as European Article Number (EAN-13), ensuring fast and accurate scanning at the point of sale.
5. Real-time Price Calculation - The system must calculate and display the price and weight of the produce in real time based on the measured weight and the unit price per kilogram. The price is calculated based on the measured weight and the predefined price per kilogram. Since the system obtains weight in grams, it is first



converted to kilograms. The formula used is: $Price =$

$$\frac{Weight\ in\ grams}{1000g} \times Price\ per\ kilogram$$

where the weight in grams is obtained from the load cell and converted to kilograms before computing the total price. Price should update instantly on the Liquid Crystal Display (LCD) after weighing, ensuring user transparency and convenience. Accurate real-time calculation also supports administrative control and prevents pricing errors during transactions.

6. Interface for an Administrator - The prototype must support administrative control through an interface that allows easy modification of produce prices, with changes reflected immediately in the real-time calculation module. This ensures adaptability to market price changes and supports operational flexibility.

Design Plan and Fabrication

This section presents the design plan and fabrication of the self-service vegetable purchase prototype, outlining its key components, functional requirements, and systematic assembly to achieve accurate, automated weighing, pricing, and packaging.

Step 1: Conceptual Research and Planning

The process begins with in-depth research on self-service vegetable checkout systems, relevant technologies, and best practices in automation and retail. The objectives and functional requirements of the prototype are defined, including supported vegetables, weighing capacity, image recognition performance, and packaging constraints.



Step 2: Feasibility Analysis and Component Research

Technical information on hardware and software components is gathered, including Raspberry Pi, load cell, camera module, heat sealers, thermal printer, and LCD Displays. Constraints such as weight capacity, lighting conditions, bag type, and sensor accuracy are evaluated. Cost, availability, and compatibility of components are also considered to ensure practical implementation.

Step 3: Procurement of Hardware and Software Components

All necessary components are sourced and acquired, including mechanical parts, electronic modules, and a microcontroller. Software tools and libraries, such as Edge Impulse, are also prepared for the prototype. Attention is given to ensuring compatibility, quality, and quantity for smooth integration.

Table 1: Components and Function Features

Components	Functions
Raspberry Pi 4 Model B	Serves as the main processing unit, controlling the entire system, running the image detection model, handling weight data, calculating prices, and coordinating the output to the LCD, thermal printer, and sealing mechanism.
Raspberry Pi Camera Module V2 (IMX219)	Captures images for the machine learning model to perform image detection and identify the type of produce (red onion, garlic, marble potato).
Load Cell	Measures the weight of the produce placed on the platform with precision. Converts the physical weight into an electrical signal that the HX711 module can read.



HX711 Module	Acts as an analog-to-digital converter for the load cell, amplifying and converting the weight signal so the Raspberry Pi can process it accurately.
Liquid Crystal Display (LCD) with i2c Module	Displays the vegetable type detected, accuracy of the vegetable detected, weight and the price of the vegetable.
Thermal Printer	Prints barcode receipts or price labels automatically and barcode for scanning at checkout.
Impulse Sealer	Seals the produce in heat-sealable LDPE plastic bags, ensuring secure, hygienic packaging and a ready-to-purchase product.
Connecting Wires	Connect the Raspberry Pi to the sensors and modules to enable communication and power between components.
Actuator	The linear actuator serves as the mechanical arm of the system that pushes and pulls the impulse sealer during the packaging process. It automates the sealing action after the user confirms the transaction.
Thermal Printer	The thermal printer prints the barcode receipt containing the vegetable name, weight, and computed price. It provides a fast and ink-free way of generating transaction output for the user.
Power Supply	The power supply provides the required electrical power for the Raspberry Pi and other hardware components of the device. It is important for maintaining stable and safe operation of the system.
Barcode Scanner	The barcode scanner reads the printed barcode generated by the system after



weighing and packaging. It is used to retrieve the saved transaction data from the database for verification or cashier-side processing.

Stepper Motor Drive

The stepper motor drive is a controller used for precise stepper motor movement. In our current device, it is not part of the main implemented setup, but it may be used in future upgrades for additional mechanical control.

Buck Converter

The buck converter steps down a higher DC voltage into a lower stable voltage needed by sensitive components. It helps safely power parts of the system while improving efficiency.

Terminal Block

The terminal block is used to organize and secure wiring connections in the device. It makes the electrical setup cleaner, safer, and easier to troubleshoot.

Raspberry Pi Flex Cable

The Raspberry Pi flex cable connects the Camera Module V2 to the Raspberry Pi through the CSI port. It allows the camera to send real-time image data for vegetable detection.

Step 4: System Design and Layout Planning

A detailed design is developed using block diagrams and schematic layout for the microcontroller. This includes the placement of mechanical structures: a load cell, a camera, a sealing mechanism, and display units. The operational sequence of the system, from image detection to weight sensing, pricing, and packaging, is defined to ensure an efficient workflow.

Step 5: Individual Module Testing and Validation



Each module is tested separately to verify proper functionality. Weight sensing is calibrated for accuracy, image detection is validated with sample produce, which are red onion, garlic, and marble potato, the LCD is tested for correct display, thermal printing is validated for accuracy and legibility, and the heat sealer is checked for consistent sealing. Any issues identified are documented and corrected before integration.

Step 6: Run Computing Simulations for Image Detection

The image detection model is trained using sample images of red onion, garlic, and marble potatoes. Simulations run under varying lighting conditions to evaluate detection accuracy and robustness. Adjustments are made to optimize detection performance and minimize errors.

Step 7: Assembly and System Integration

Mechanical components are assembled, and electronic modules are mounted according to the design layout. All software and hardware components are integrated, including the load cell, camera, LCD, sealing mechanism, and database. The system is configured to operate sequentially, ensuring smooth interaction between subsystems.

Step 8: System Testing and Performance Evaluation

The fully assembled prototype is tested under realistic conditions to evaluate end-to-end functionality. Performance metrics such as weight measurement accuracy, image detection success rate, price calculation correctness, and sealing reliability are recorded. Necessary adjustments and troubleshooting are performed to optimize system performance.

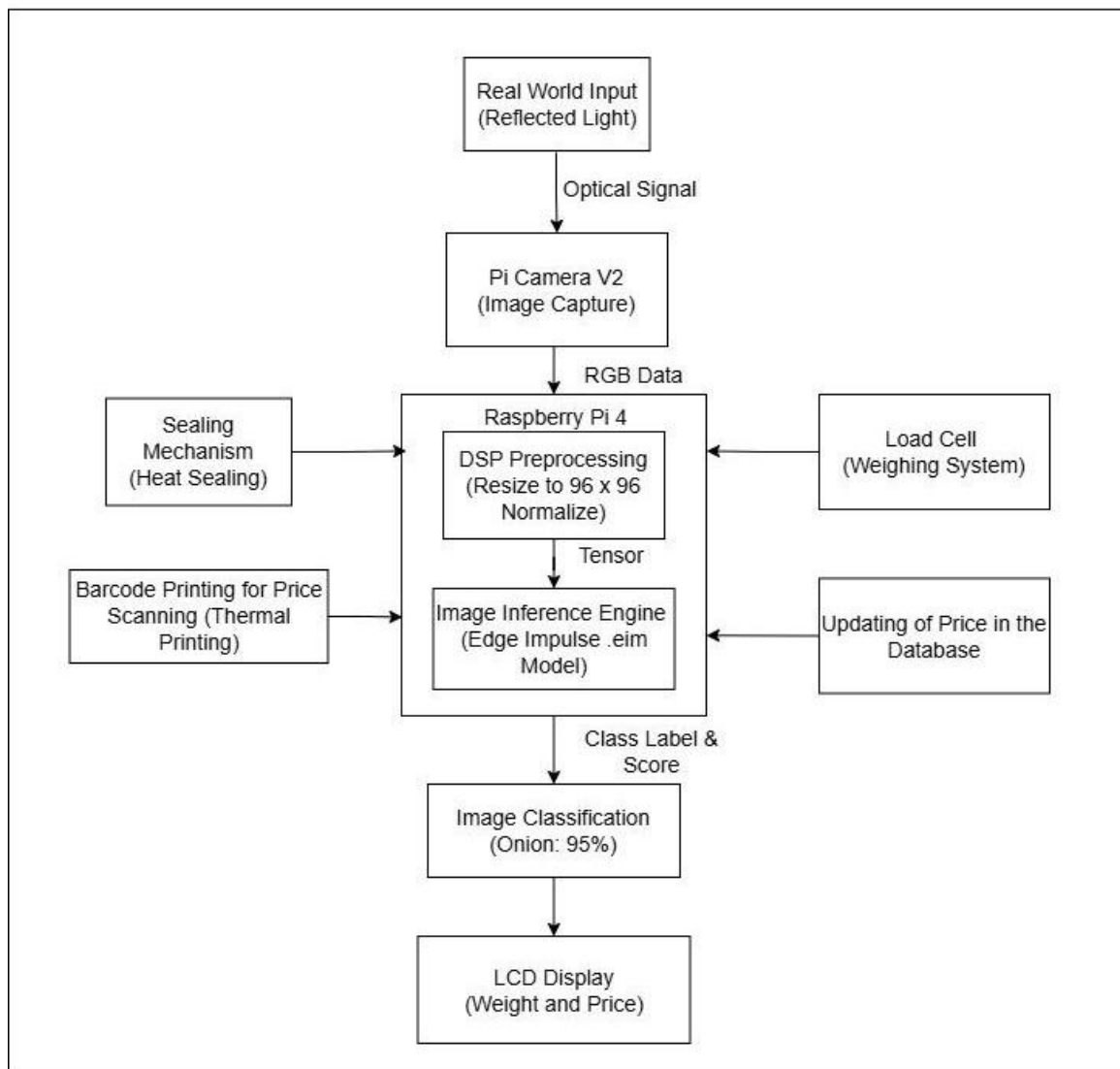


Step 9: Evaluation and Documentation of Results

The final results are analyzed and compared against the design criteria, including limitations identified during testing. Observations on accuracy, reliability, and usability are documented. In addition, a self-made questionnaire was administered to gather user feedback on the system's ease of use, interface clarity, speed of operation, and overall satisfaction. The responses are analyzed to evaluate user experience and inform improvements. Recommendations for future enhancements and potential scalability are also provided to guide future development.



Figure 2. Block Diagram





The block diagram represents the functional architecture of the automated produce identification and weighing system. Optical data captured by the Raspberry Pi Camera Module V2 is converted into RGB image data and transmitted to the Raspberry Pi 4, which functions as the central processing unit. In parallel, a load cell with HX711 amplifier measures the weight of the produce and provides digital weight data to the same processing unit.

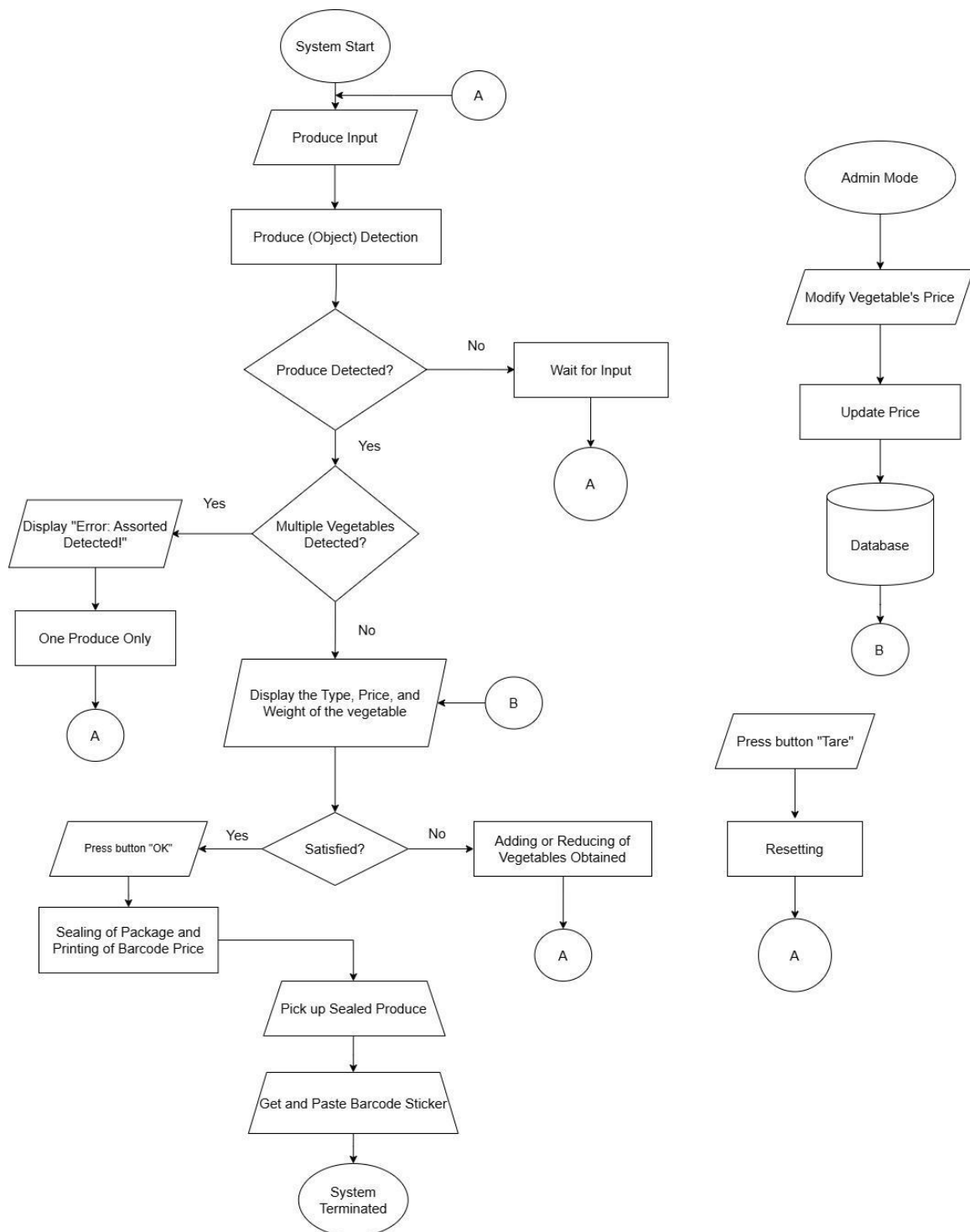
Within the Raspberry Pi, the image data undergoes preprocessing, including resizing to 160 x 160 pixels and normalization, before being converted into a Edge Impulse executable (.eim) format suitable for inference. The processed data are analyzed by a Edge Impulse inference engine executing a Faster Objects, More Objects (FOMO)-based object detection model, which outputs a predicted class label and confidence score (0 - 100%). The classification result is then linked to a locally stored price database, where each recognized vegetable has an assigned unit price per kilogram.

Weight data from the load cell and price data retrieved from the database are combined to compute the total cost of the produce in real time. The type of vegetable and its accuracy, calculated weight and price are transmitted to the LCD module for user display. Transaction data are simultaneously forwarded to a thermal barcode printer, which generates a scannable barcode encoding the item details, weight, and price. Packaging operations are handled through a heat-sealing mechanism controlled by the Raspberry Pi.



An administrator interface is integrated into the system, allowing authorized personnel to update vegetable prices stored in the local database. Any price modification made through this database is immediately reflected in the real-time pricing calculations without requiring system recalibration or retraining of the image classification model. All system components are powered by a regulated 5V power supply (Raspberry Pi, 2019), ensuring stable and continuous operation of the automated checkout system.

Figure 3. System Flowchart





The flowchart illustrates the complete operational logic of the proposed self-service vegetable purchase system, consisting of two coordinated processes: the customer transaction flow and the administrative price update flow.

The process begins when the system initializes all required modules, including the camera, load cell, display, and database connection. The customer then places a vegetable on the weighing platform. The camera captures an image of the produce, and the system performs image processing and detection to identify the vegetable type.

At this stage, the system evaluates the detection result. If no vegetable is detected (empty platform), the system will wait for a valid input, looping back to the image processing stage without processing to weighing or pricing. If the user wants to reset the system, the user can press the button “Tare.”

If multiple vegetables or mixed produce are detected simultaneously, the system triggers an error state, displaying an “Error: Assorted Detected” message and optionally activating an alarm indicator. This mechanism prevents improper transactions and ensures that only one type of vegetable is processed at a time. The system remains in this error loop until the excess product is removed and a valid single-vegetable input is detected.

Once the vegetable is successfully identified, the system displays the type and accuracy rate of the vegetable, and its price and weight. At this stage, the load cell measures the produce weight, while the Raspberry Pi retrieves the latest unit price of the identified vegetable from the local database. The system then calculates the total price



based on the measured weight and displays both the weight and price on the LCD in real time.

The system next evaluates user confirmation. If the user is dissatisfied with the quantity of produce, the system detects changes in the input and returns to the image processing stage, allowing re-identification, re-weighing, and price recalculation. This loop continues until the user confirms the transaction details.

If the user is satisfied with the product quantity, the transaction is finalized by pressing the OK button. The system then activates the sealing mechanism to securely package the produce and initiates barcode printing, generating a barcode sticker containing the details for price scanning. The sealed product is released to the user.

In parallel, the administrative price management flow allows authorized personnel to update vegetable prices. The administrator inputs revised pricing values. The updated prices are stored in the local database and are accessed during customer transactions, ensuring that all price computations reflect the most current values without interrupting system operation.



Plan of Implementation

Table 2. Time Table

Task	Oct. 2025	Nov. 2025	Dec. 2025	Jan. 2026	Feb. 2026	Mar. 2026	Apr. 2026
Research and Planning of the Study							
Consultation of the Study							
Documenting Chapter 1							
Researching and Gathering of Materials and Components							
Researching Related Literatures and Related Studies							
Constructing Chapter 2							
Testing the Components							
Image Detection Simulation (1 Produce)							
Chapter 3 Documentation							
Image Detection Simulation (Remaining Produce)							
Database and Interface for the Admin							
Barcode Printing Integration							
Constructing the Sealing Mechanism							
Testing the Device							
Evaluation to the Respondents							
Documenting Chapter 4							
Documenting Chapter 5							

The Gantt chart outlines an eight-month project timeline from October 2025 to April 2026, structured into four main phases. The Research and Planning phase focuses on the literature review, consultations, and documentation for Chapter 1. The Development and Initial Testing phase involves component testing, image detection simulation, and Chapter 2 documentation. The Integration and System Completion phase covers system assembly, database and admin interface development, barcode printer integration, and sealing mechanism construction. Finally, the Evaluation and Documentation phase includes system testing, user evaluation, and completion of Chapters 4 and 5. This timeline ensures a systematic progression from development to validation of the proposed system.



Cost of Materials

Table 3. Bill of Materials

Components			Quantity (by piece)	Unit Cost (Php)	Total (Php)
Raspberry Pi 4 Model B			1	6,175.00	6,175.00
Load Cell with HX711 Module			1	435.00	435.00
Raspberry Pi Camera Module V2 (IMX219)			1	1,935.00	1,935.00
Liquid Crystal Display with i2C Module (LCD)			1	70.00	70.00
Push Button			2	13.00	26.00
Resistor (330 Ω)			1	2.00	2.00
Linear Actuator			1	1,680.00	1,680.00
Heat Impulse Sealer			1	580.00	580.00
Power Supply			1	415.00	415.00
Barrel Jack			1	130.00	130.00



Stepper Drive	Motor	1	340.00	340.00
Buck Converter		1	85.00	85.00
Terminal (Set)	Block	20	44.00 (10 pieces)	88.00
RPi Flex Cable (1000mm)		1	130.00	130.00
Thermal Printer		1	1,595.00	1,595.00
Thermal Sticker		1	210.00	210.00
Barcode Scanner		1	870.00	870.00
Angle (Stainless)	Bar	15	5.00	205.00
Flat Head Screw (Set)		48	205.00	205.00
Acrylic Casing		1	3,600.00	3,600.00
Plywood (10mm)		1	600.00	600.00
Pulley		2	17.50	35.00
Total				19,411.00



Table 1 presents the list of hardware components used in the development of the prototype, including their quantities, unit costs, and total costs. The total cost of materials for the prototype is ₱19,411.00, demonstrating that the system can be implemented using affordable, readily available components.

Evaluation Procedure

The assessment of the smart produce identification system was conducted through a methodical, dual-stage approach to verify its technical operation and evaluate its interaction with users. The initial stage involves rigorous technical verification under laboratory conditions, where component accuracy is confirmed using precision weights and labeled image sets, and the full system's functionality is examined to determine transaction reliability, processing time, and pricing accuracy. This stage also intentionally assesses specified system responses, such as activating error alerts for multiple items, entering standby mode when no produce is detected, and ensuring real-time updates from the administrative pricing database are applied correctly. In the subsequent stage, hands-on, scenario-based trials are carried out with intended users in a simulated setting.

After completing practical tasks, participants provide feedback through a standardized survey that includes a 5-point Likert scale and open-ended items to gauge ease of use, perceived dependability, and overall satisfaction. The collected data, both measurable outcomes from technical trials and subjective input from user evaluations, are then combined and examined using statistical summaries and thematic analysis,



providing a well-rounded appraisal that connects engineering performance with practical user acceptance.

Instrumentation

A structured self-made questionnaire was designed to serve as the primary data collection instrument for this study. It consists of three main sections: demographic profile, core construct assessment, and open-ended feedback. The core assessment section utilizes a 5-point Likert scale (ranging from 1=Strongly Disagree to 5=Strongly Agree) to measure respondent perceptions across key dimensions relevant to the research objectives (Kusmaryono et al., 2022). This includes image detection, weight sensing, sealing mechanism, real-time price calculation, and an interface for an administrator. The final section provides space for optional qualitative comments. The instrument was formatted for clarity and ease of administration, ensuring respondents could complete it in a standardized manner.

Validity and Reliability of Research Instrument

The questionnaire used in this study was subjected to expert validation to ensure its content validity and appropriateness in evaluating the developed system. A total of five (5) validators, composed of faculty members with expertise in engineering and system development, reviewed the instrument. Their feedback was utilized to improve the clarity, relevance, and organization of the questionnaire items. Using a 5-point Likert



scale, the instrument obtained an overall mean rating of 4.51, which exceeded the acceptable threshold of 4.5. This result confirmed that the questionnaire possessed strong content validity (Almanasreh et al., 2022).

In addition, a pilot test was conducted involving ten (10) participants to assess the reliability of the questionnaire. The collected responses were analyzed using Cronbach's Alpha to determine the internal consistency of the instrument. The resulting coefficient of 0.938 indicated a high level of reliability, confirming that the instrument was appropriate for gathering data (Cheung et al., 2024).

After validation and reliability testing, the finalized questionnaire was administered to thirty (30) respondents for the evaluation of the system in terms of usability, functionality, and reliability.

Data to be Gathered

The study gathered data from 30 respondents, comprising students, teachers, vendors, and consumers. This diverse group was chosen to capture perspectives from end-users, represented by students, and supervisory or administrative users, represented by teachers. The data included demographic information such as role, and relevant experience with technology or automated systems. In addition, the study collected information on perceptions and satisfaction with the system's key functionalities, including image detection accuracy, weight-sensing precision, sealing mechanism efficiency, real-time price calculation, and the usability of the administrator interface.



Finally, qualitative feedback was obtained through open-ended questions that allow respondents to provide suggestions, observations, or concerns about the system's performance and user experience. The combination of quantitative and qualitative data from these 30 participants providing a comprehensive understanding of the system's usability, effectiveness, and overall acceptance within the community

Data Analysis

The data collected from the 30 respondents were analyzed using both quantitative and qualitative methods. Quantitative responses from the Likert-scale questionnaire were summarized using descriptive statistics, including frequencies, percentages, means, and standard deviations. These were used to determine trends and evaluate respondents' level of satisfaction with the system's functionalities, such as image detection, weight sensing, sealing mechanism, real-time price calculation, and administrator interface usability. The mean was used to determine the overall evaluation of the respondents, while the standard deviation measured the consistency and variation of their answers.

Qualitative responses from the open-ended questions were analyzed using thematic analysis to identify recurring ideas, suggestions, and concerns expressed by the respondents. Combining quantitative and qualitative methods provided a comprehensive understanding of the system's usability, functionality, reliability, effectiveness, and overall acceptability among students, teachers, vendors, and consumers.



Table 4. Mean Score Interpretation for Prototype Evaluation

Mean Scale	Interpretation	Description
4.21 – 5.00	Very High	Respondents strongly agree; excellent perception of the system.
3.41 – 4.20	High	Respondents agree; above-average perception of the system.
2.61 – 3.40	Moderate	Respondents are neutral or somewhat agree; average perception of the system.
1.81 – 2.60	Low	Respondents disagree; below average perception of the system.
1.0 - 1.80	Very Low	Respondents strongly disagree; very poor perception of the system.



Parameters for Data Analysis

Table 5. Parameters for Data Analysis

Parameter	What it measures	How computed	Typical Evidence
Classification Accuracy (%)	Detection correctness	$(\text{Correct predictions} / \text{Total predictions}) \times 100$	Confusion matrix / validation logs
Assorted Detection Accuracy (%)	Ability to detect multiple vegetables simultaneously	$(\text{Correct multi-object detections} / \text{Total test cases}) \times 100$	Detection logs / annotated test images
Weight Measurement Error (grams)	Deviation from true weight	$ \text{Measured Weight} - \text{Actual Weight} $	Calibration logs / repeated trials
Price Computation Accuracy (%)	Correctness of calculated price	$(\text{Correct price calculations} / \text{Total transactions}) \times 100$	Transaction logs / manual verification
Average System Response Time (seconds)	System speed	$(\text{Sum of response times}) / \text{Number of requests}$	Performance logs/timestamp analysis
Barcode and Printing Accuracy (%)	Correctness of barcode output and print results	$(\text{Correct prints} / \text{Total prints}) \times 100$	Printed receipts / scan verification
Sealing Mechanism Reliability (%)	Success rate of sealing operation	$(\text{Successful seals} / \text{Total attempts})$	Trial logs / observation records



x100

System Stability Across Repeated Trials	Consistency under repeated use	(Successful trials / Total Trials) x 100	Trial logs/ uptime records/ error reports
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Table 5 outlines key performance parameters used for data analysis, focusing on the accuracy, precision, speed, and reliability of a system. Detection Accuracy measures the system's ability to correctly detect or categorize data. It is calculated by dividing the number of correct predictions by the total number of predictions, then multiplying by 100. This is typically supported by confusion matrices or validation logs. Weight Measurement Error captures how much the system's weight readings deviate from actual values, computed as the absolute difference between the measured and actual weights. Calibration logs or repeated trials often serve as supporting evidence. Price Computation Accuracy evaluates the correctness of calculated prices during transactions and is determined by dividing correct computations by total transactions and multiplying by 100. It is validated through transaction logs or manual records. Average System Response Time gauges how quickly the system responds to requests, derived from dividing the total response time by the number of requests. This parameter is backed by timestamp analyses. System Stability Across Repeated Trials reflects the consistency of the system's performance over multiple uses, calculated as the percentage of successful trials out of total trials, with evidence from uptime records or error logs.



Chapter 4

RESULTS AND DISCUSSION

System Output

The system output demonstrates the complete functionality of the prototype, including vegetable detection, weight measurement, price computation, barcode generation, and sealing. The following figures present the actual outputs of the system during operation.

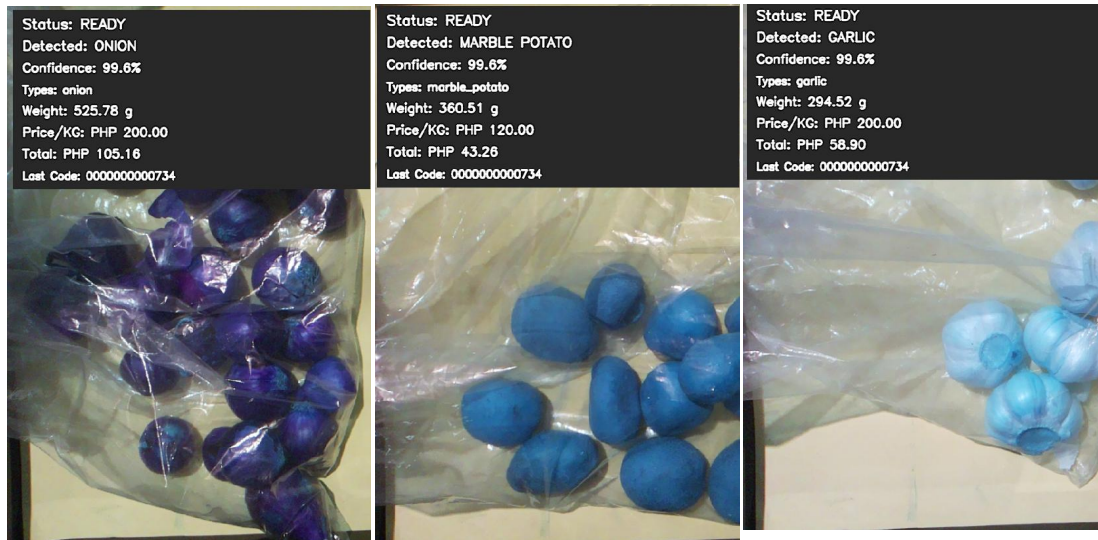


Figure 4. Vegetable Detection

Figure 4 shows that the vegetables detected successfully the three (3) vegetables when placed in the platform. The interface displays the detected vegetable type along with a high confidence level, indicating accurate classification. Additionally, the system shows the measured weight and computed price, confirming proper integration of the image detection, weight sensing, and price computation modules.



Figure 5. Liquid Crystal Display (LCD) User Interface



The LCD interface displays real-time information including the detected vegetable type, measured weight, and computed price. This provides immediate feedback to the user, ensuring transparency, and usability during the transaction process.

Figure 6. Printed Barcode



As shown in Figure 7, the printed barcode contains transaction details such as vegetable type, weight, total price, and a unique identification code. The barcode is clearly printed and readable, demonstrating the system's capability to generate reliable output for transaction tracking and verification. The generated barcode is intended to be scanned using a barcode scanner for efficient data retrieval.



Figure 7. Sealed Product



The sealed product demonstrates the final stage of the system, where the packaging is secured using the sealing mechanism. The output confirms that the system is capable of completing the full process from detection to packaging.

Trials and Results of Technical Features

Classification Accuracy

Table 6. Evaluation in Image Detection for Red Onion

Trial	Actual	Predicted	Correct
1	Onion	Onion	✓
2	Onion	Onion	✓
3	Onion	Onion	✓
4	Onion	Onion	✓



5 Onion Onion ✓

Table 7. Evaluation in Image Detection for Garlic

Trial	Actual	Predicted	Correct
1	Garlic	Garlic	✓
2	Garlic	Garlic	✓
3	Garlic	Garlic	✓
4	Garlic	Marble Potato	✗
5	Garlic	Garlic	✓

Table 8. Evaluation in Image Detection for Marble Potato

Trial	Actual	Predicted	Correct
1	Marble Potato	Marble Potato	✓
2	Marble Potato	Marble Potato	✓
3	Marble Potato	Garlic	✗
4	Marble Potato	Marble Potato	✓
5	Marble Potato	Onion	✗

Table 6, 7, 8 shows the results of every trials of each vegetable. The system performs differently across the three vegetable types tested. The red onion achieved 100% accuracy across all trials, indicating that the model is highly effective in recognizing this vegetable. Garlic obtained 80% accuracy, with only one



misclassification observed, while marble potato showed a lower accuracy of 60%, with multiple incorrect predictions. These results suggest that the model performs well when distinguishing vegetables with unique visual features but encounters difficulty when processing objects with similar characteristics.

The variation in classification performance can be attributed to the visual similarities between garlic and marble potato, such as size, shape, and color, which may confuse the model during prediction. In contrast, the red onion has more distinct features that allow for consistent recognition. These findings indicate that while the model is generally reliable, improvements such as increasing dataset diversity, enhancing image quality, and refining training parameters may help improve accuracy, particularly for marble potato detection.



Assorted Detection Accuracy Assorted Vegetables Detected

Table 9. Evaluation in Image Detection for Assorted Vegetables

Trial	Objects Present	Detected	Correct
1	Marble Potato + Onion	Assorted Vegetables Detected	✓
2	Onion + Garlic	Assorted Vegetables Detected	✓
3	Garlic + Marble Potato	Assorted Vegetables Detected	✓
4	Marble Potato + Onion + Garlic	Assorted Vegetables Detected	✓
5	Onion + Marble Potato + Garlic	Assorted Vegetables Detected	✓

Table 9 presents the multiple trials tested by the system in terms of assorted vegetables. The system achieved 100% accuracy in detecting assorted vegetables across all trials. In each case where multiple vegetable types were placed on the platform, the system correctly identified the presence of mixed items and triggered the appropriate



response. This confirms that the system is capable of recognizing complex input scenarios involving more than one object.

The consistent performance in multi-object detection is a critical feature of the system, as it prevents incorrect transactions by ensuring that only one type of vegetable is processed at a time. This contributes to maintaining transaction integrity and system reliability. The results indicate that the object detection model is effective in distinguishing between single and multiple object conditions, even when different vegetables are placed together.

Weight Measurement Accuracy and Precision

Table 10. Evaluation in Weight Measured Accuracy

Trial	Actual (g)	Measured (g)	Error
1	100	99.7	0.3
2	200	194.4	5.6
3	80	79.9	0.1
4	135	134.6	0.4
5	889	888.1	1.1

As shown in Table 10 is the trials for the weighing mechanism of the prototype. The weight measurement results indicate that the system provides generally accurate readings, with most trials falling within the acceptable tolerance range of ± 5 grams. The



recorded errors were minimal in most cases, demonstrating the capability of the load cell system to produce consistent measurements suitable for practical use. However, one trial showed a higher deviation, slightly exceeding the acceptable tolerance.

This deviation may be attributed to factors such as improper placement of the object on the weighing platform, uneven distribution of weight, or calibration limitations of the load cell. Despite this, the overall performance of the weight measurement system remains reliable. The results suggest that while the system is sufficiently accurate for price computation, further calibration and mechanical stabilization could improve precision and consistency.

Price Computation Accuracy

Table 11. Evaluation in Price Computation

Trial	Expected Price (₱)	System Price (₱)	Correct
1	164.52	164.52	✓
2	53.0628	53.06	✓
3	52.142	52.14	✓
4	3.252	3.25	✓
5	51.476	51.48	✓

Table 11 shows the computation capabilities of the system. The system achieved 100% accuracy in price computation across all trials when values are rounded to two



decimal places. All computed prices matched the expected values after rounding, confirming that the algorithm correctly processes the measured weight and applies the corresponding unit price. This indicates that the computation module functions correctly and produces consistent and reliable results within standard currency precision.

The accuracy of price computation is essential for maintaining financial integrity in the system. Since pricing directly affects user transactions, the consistency observed in all trials ensures reliability and trustworthiness of the system. These results confirm that the software logic used for calculation is correctly implemented and does not introduce computational errors.

Barcode Printing Accuracy

Table 12. Evaluation in Barcode Printing

Trial	Barcode Generated	Printed Clearly	Scanned Successfully	Match with Database	Result
1	0000000000680	Yes	Yes	Yes	✓
2	0000000000673	Yes	Yes	Yes	✓
3	0000000000697	Yes	Yes	Yes	✓
4	0000000000826	Yes	Yes	Yes	✓
5	0000000000833	Yes	Yes	Yes	✓

Table 12 presents the barcode printing of the system. The barcode generation and printing system achieved 100% accuracy in all trials. Each generated barcode was printed



clearly, successfully scanned, and correctly matched with the corresponding database record. This demonstrates that the system can reliably produce valid and readable barcodes for transaction identification.

The effectiveness of the barcode system ensures that transaction data can be efficiently retrieved and verified. The clear print quality and consistent scanning performance indicate proper integration between the printer, barcode generation logic, and database system. These results confirm that the barcode module is highly reliable and suitable for practical implementation in automated systems.

Sealing Reliability

Table 13. Evaluation in Sealing Reliability

Trial	Seal Success	Remarks
1	No	Misalignment of plastic bag
2	No	Incomplete seal (low heat)
3	Yes	Proper seal, no leakage
4	Yes	Proper seal, no leakage
5	Yes	Strong and consistent seal

Table 13 shows the sealing capabilities and reliability of the prototype. The sealing mechanism achieved a success rate of 60%, with three successful trials and two failures. The successful trials produced properly sealed packages with no leakage, while



the failed trials were due to issues such as misalignment of the plastic bag and insufficient heat during the sealing process.

These results indicate that while the sealing mechanism is capable of performing its intended function, its reliability is not yet consistent. The failures highlight the need for improved alignment control and better heat regulation. Enhancing these aspects would improve the overall performance and ensure consistent sealing results in future implementations.

System Stability

Table 14. Evaluation in Consistency of the Prototype

Trial	System Status	Error Occurred	Remarks
1	Failed	Yes	Low heat seal
2	Success	No	Stable
3	Success	No	Stable
4	Success	No	Stable
5	Success	No	Stable

As presented in Table 14, the prototype demonstrated an overall stability rate of 80%, with most trials completing successfully without errors. Only one trial resulted in failure, which was primarily caused by issues related to the sealing mechanism. The majority of system operations, including detection, weighing, computation, and printing, functioned as expected.



The results indicate that the system is generally stable and capable of operating under repeated use. However, the observed failure suggests that system stability is influenced by the reliability of individual hardware components, particularly the sealing module. Improving the performance of these components would enhance overall system stability and ensure consistent operation.

Development of User's Manual

User's Manual

The User's Manual serves as a guide for the proper operation of the system, covering its key functions and features. The complete discussion and presentation of the user's manual is shown in Appendix G.

Figure 8. User's Manual Cover Page



AUTOMATED PRODUCE SCANNER WITH REAL-TIME PRICING AND PACKAGING

User Manual



BRILLAS | GONZALES | JAMELANO | LACSON | MAGAN

MAY 2026 VERSION 1.0

Data Gathering Procedures

The data for this study were collected using a self-made questionnaire developed by the researchers to evaluate the performance and acceptability of the proposed system. The instrument was designed to assess the system in terms of usability, functionality, and reliability.



Prior to the actual data collection, the questionnaire was subjected to expert validation by five (5) qualified validators to ensure the clarity, relevance, and appropriateness of each item. Revisions were made based on their feedback to improve the quality of the instrument.

A pilot test was also conducted involving ten (10) participants to assess the reliability of the questionnaire. The collected data were analyzed using Cronbach's Alpha, which yielded a value of 0.977, indicating excellent internal consistency of the instrument.

After validation and reliability testing, the finalized questionnaire was distributed to thirty (30) respondents, composed of students and individuals relevant to the system's intended use. Their responses were then collected, tabulated, and analyzed to evaluate the overall performance and acceptability of the developed prototype.

Results of the Parameters of Evaluation of the Prototype

To validate the feasibility of the prototype in terms of usability, functionality, and reliability, the researchers used a self-made validated questionnaire to survey 30 respondents. The respondents will be provided with a survey via Google Form depending on the availability, and they will be given a short period of time.

Parameters and Interpretation



Table 15. Evaluation results of students, teachers, consumers, and vendors using self-made questionnaire

Parameters	Mean	Standard Deviation	Interpretation
Usability	4.66	0.046	Very High
Functionality	4.63	0.098	Very High
Reliability	4.65	0.092	Very High
Overall	4.65	0.08	Very High

Table 15 presents the evaluation results of the prototype based on the responses of the participants. The system obtained a mean rating of 4.66 for usability, indicating that it is highly user-friendly and easy to operate. In terms of functionality, the system achieved a mean of 4.63, reflecting that it performs its intended operations accurately and effectively. Meanwhile, reliability obtained a mean of 4.65, indicating that the system operates consistently with minimal errors.

The standard deviation values for usability (0.046), functionality (0.098), and reliability (0.092) indicate a very high level of consistency among the responses. These low values show that the respondents strongly agree in their evaluation of the system.

Overall, the results suggest that the developed prototype is highly acceptable and demonstrates excellent performance in terms of usability, functionality, and reliability. Furthermore, Cronbach's Alpha value confirms that the evaluation instrument used in this study is highly reliable and consistent.



Chapter 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter discusses the summarized findings of the study, the conclusions derived from the results, and the recommendations proposed by the proponents.

Summary of Findings

This study aimed to design and develop an automated vegetable classification, weighing, and labeling system integrating computer vision, load cell measurement, and electromechanical components. The findings revealed that the system effectively performed real-time image detection and classification using the trained Edge Impulse



model, successfully identifying red onion, garlic, and marble potato under operational conditions. Notably, the system achieved 100% accuracy in detecting assorted vegetables, ensuring that mixed items were properly identified and prevented from being processed, thereby maintaining transaction integrity. However, the overall classification accuracy reached approximately 80%, with performance varying depending on the vegetable type.

In terms of weight measurement, the load cell system demonstrated reliable performance, with most readings falling within an acceptable tolerance range of ± 5 grams compared to actual values. Although minor fluctuations were observed during testing, the system maintained consistent output, indicating its suitability for practical use. Furthermore, the system achieved 100% accuracy in real-time price computation, with calculated prices consistently matching the expected values based on the measured weight and predefined price per kilogram.

The barcode generation and printing functionality also exhibited complete accuracy, as all generated barcodes were printed clearly, successfully scanned, and correctly matched with the corresponding records stored in the database. Meanwhile, the automated sealing mechanism initially encountered inconsistencies due to factors such as improper alignment and insufficient heat; however, after calibration, the system demonstrated reliable and consistent sealing performance. Overall, the system exhibited stable operation during continuous use, with minimal errors after adjustments. It was also observed that classification accuracy varied depending on the vegetable type, with marble



potato showing lower consistency compared to onion, indicating the need for further model improvement.

Conclusions

Based on the results of the study, it can be concluded that the developed automated vegetable classification, weighing, and labeling system is functional, reliable, and effective in achieving its intended objectives. The integration of computer vision and load cell technology successfully enabled accurate identification and measurement of vegetables within a self-service environment. The system's ability to detect assorted vegetables with perfect accuracy plays a crucial role in ensuring transaction integrity by preventing the processing of mixed items.

Moreover, the system demonstrated excellent performance in terms of price computation and barcode generation, both of which achieved complete accuracy and reliability, ensuring correct and traceable transactions. Although the weighing system exhibited minor fluctuations, the results remained within an acceptable range, making it suitable for real-world applications. The automated sealing mechanism, despite initial inconsistencies, proved to be reliable after proper calibration, highlighting the importance of hardware optimization in achieving stable performance. Overall, the study confirms that low-cost embedded systems such as the Raspberry Pi, combined with machine



learning models from Edge Impulse, can be effectively utilized to develop efficient, accurate, and practical automated retail solutions.

Recommendations

Based on the findings and identified limitations of the study, several recommendations are proposed to further enhance the system's performance. It is recommended to improve the image detection model by increasing the dataset size, particularly for marble potato, and incorporating more diverse training images that include variations in lighting conditions, object positioning, and background environments. Enhancing the quality and consistency of labeled data is also essential to improve model accuracy.

In addition, implementing a controlled lighting system, such as the use of LED illumination or an enclosed imaging environment, is highly recommended to minimize the impact of environmental lighting variations on image detection performance. For the weighing system, the application of filtering techniques, such as a moving average or noise reduction algorithm, is suggested to further stabilize weight readings and reduce fluctuations.

Furthermore, future improvements may include expanding the system's capability to recognize a wider variety of vegetables, enhancing hardware components such as the



sealing mechanism for better alignment and heat control, and integrating additional features such as cloud-based data storage or multi-terminal support. Improving the user interface through clearer prompts or feedback mechanisms is also recommended to enhance user experience and operational efficiency.

The researchers also recommend the continued development of automated retail technologies that support Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure by promoting technological innovation and automation, and SDG 12: Responsible Consumption and Production by improving operational efficiency, reducing human error, and supporting smarter and more sustainable retail practices (Hariyani et al., 2025).

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APPENDIX A

SURVEY QUESTIONNAIRE



Carlos Hilado Memorial State University

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Questionnaire for Automated Produce Scanner with Real-time Pricing and Packaging

Respondent Profile

Name (Optional): _____

Category/ Sector:

Student ☐

Teacher/ Faculty ☐

Vendor ☐

Consumer ☐

Do you have an experience with automated systems or self-checkout kiosks?

Yes ☐

No ☐

How often do you often purchase in supermarkets?

Daily ☐

Weekly ☐

Monthly ☐

Rarely ☐

Never ☐



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Instructions:

Please rate your level of agreement with each statement by checking (✓) the corresponding number based on the following scale:

Rating	Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

A. Hardware Specification Questions

This section assesses the hardware specifications of the device features.

Statement	5	4	3	2	1
1. The developed weighing scale provides accurate weight measurements for the vegetables compared to standard weights.					
2. The weight reading on the Liquid Crystal Display (LCD) displays updates real-time when vegetables are added or removed.					
3. The tare (zeroing) function to reset works properly before weighing.					
4. The prototype can handle weights up to 3 kilograms without issues.					
5. The sealing mechanism securely closes the opening of the plastic bag containing the vegetables.					
6. The sealing process is activated properly when the OK button is pressed.					



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7. The sealed package is neat and prevents spillage of the vegetables.					
8. The system works correctly with the provided heat-sealable Low-Density Polyethylene (LDPE) plastic bags.					
9. The sealing process is completed within a reasonable response time.					
10. The thermal printer produces a clear and readable barcode sticker.					
11. The printed barcode contains the transaction information (item, weight, and price).					
12. The printing of price process is completed after transaction confirmation.					

B. Software Specification Questions

This section assesses the software specifications of the device features.

Statement	5	4	3	2	1
1. The system accurately identifies red onion, garlic, and marble potatoes when placed on the platform.					
2. The image detection works correctly under daylight white light conditions.					
3. The system properly displays an error message when multiple vegetables are placed together.					
4. The system correctly detects when no vegetable is present on the platform.					
5. The total price displayed is correctly computed based on the vegetable type and weight.					
6. The price updates instantly when the weight changes.					
7. The price per kilogram for each vegetable is clearly displayed in					



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the Liquid Crystal Display (LCD).					
8. The final price matches the amount to be paid at the checkout counter.					
9. The real-time price display provides informed decisions about the purchase quantity.					
10. The Liquid Crystal Display (LCD) clearly shows the detected vegetable type and accuracy.					
11. The weight and price information on the display is clear.					
12. Multiple vegetables detected by the system is clearly presented in the Liquid Crystal Display (LCD) showing "Error: Assorted Detected!"					
13. The sequence of steps in using the device is easy to follow.					
14. The administrator interface allows updating of the vegetable prices.					

I. Open-Ended Feedback (Optional)

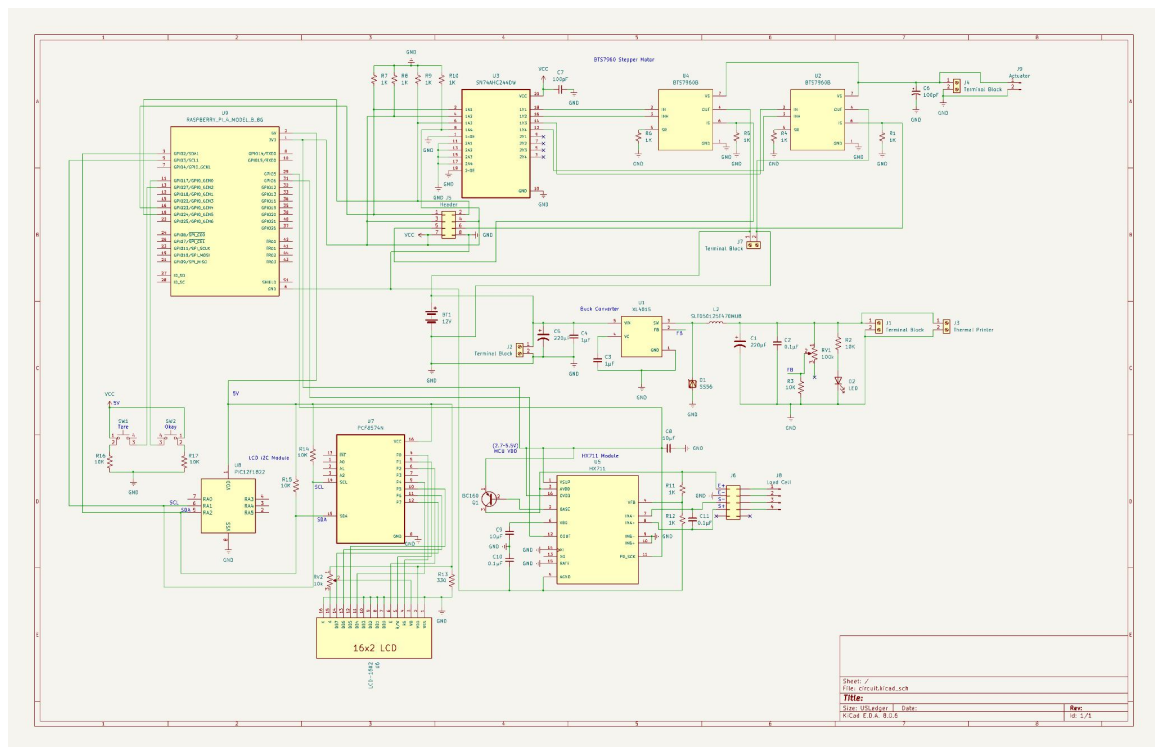
1. What phase of the system you want to improve?

2. Would you recommend this system for use in local supermarkets? Why or why not?



APPENDIX B

SCHEMATIC DIAGRAM





APPENDIX C

SOURCE CODE

```
1 #!/usr/bin/env python3
2 """
3 Integrated Raspberry Pi 4 thesis device
4 .....
5 Features:
6 - Edge Impulse image classification / FOMO object detection (.wim)
7 - Picamera2 live preview
8 - HX711 live weight reading with rolling trimmed-average stable display filter
9 - Automatic startup soft tare so the displayed/net weight starts at 0.0 g every run
10 - Automatic near-zero correction: if stable weight is between -1.0 g and -1.0 g, net display returns to 0.0 g
11 - 95% high-confidence vision logic with fallback to most-frequent low-confidence prediction
12 - High-confidence asserted-vegetable error if two or more vegetable types are strongly detected
13 - Tare button on BCM GPIO27 / physical pin 13
14 - Soft tare by short press, hard zero/reset by long press
15 - 8-kg load-cell warning/overload protection
16 - 12C 16x2 LCD display
17 - Active-HIGH confirmation button on GPIO17
18 - BTS7960-driven linear actuator sealing cycle
19 - Adjustable actuator default position before button press
20 - USB thermal printer receipt - EAN13 barcode
21 - SQLite offline storage:
22   .. 1) vegetable_price.db
23   .. 2) customer_transactions.db
24
25 Confirmed hardware mapping:
26 - Button: GPIO17, active-HIGH
27 - BTS7960 PWM: GPIO23
28 - BTS7960 LPWM: GPIO24
29 - HX711 DT: GPIO6
30 - HX711 SCK: GPIO5
31 - LCD 12C: SDA GPIO2, SCL GPIO3, address 0x27
32 - Printer: /dev/usb/lp0
33 - Tare button: GPIO27 / physical pin 13, active-HIGH
34 """
35
36 from __future__ import annotations
37
38 import json
39 import sqlite3
40 import statistics
41 import time
42 from collections import Counter, deque
43 from dataclasses import dataclass, field
44 from datetime import datetime
45 from pathlib import Path
46 from threading import Lock, Thread
47 from typing import Optional
48
49 import cv2
50 import numpy as np
51 from escpos.printer import File
52 from gpiozero import Button, DigitalOutputDevice
53 from picamera2 import Picamera2
54 from edge_impulse_linux.image import ImageImpulseRunner
55 from HX711 import SimpleHX711, Mass, Options, ReadType, Value
56 from RPLCD.i2c import CharLCD
57
58 # =====
59 # FILE PATHS
60 # =====
```




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```
start_thesis_device.sh x transaction_gui_crud.py x thesis_device_main.py x jekaca_pos_system_gui.py x thesis_device_main (6).py x
61 # =====
62 BASE_DIR = Path(__file__).resolve().parent
63 MODEL_PATH = "models/thesis_device_model_final_from_linux_armv8m-v8-isp00a-02.kie"
64 # Primary calibration file from your latest LBN big moving-average scale code.
65 # If this file is not found, the program automatically falls back to the old
66 # h712_calibration.json beside this script.
67 CAL_FILE_PATH = "models/calibration/real_environment/real_environment_test_data/h712_lbn_average_filter_calibration.json"
68 TRANS_CAL_FILE = BASE_DIR / "h712_calibration.json"
69
70 TRX_DB_PATH = "models/thesis_device/transaction_transactions.db"
71 VER_DB_PATH = "models/thesis_device/vegetable_price.db"
72 # TRX_DB_PATH = BASE_DIR / "vegetable_price.db"
73 # TRX_DB_PATH = BASE_DIR / "customer_transactions.db"
74
75 # =====
76 # CAMERA / MODEL
77 # =====
78 MODEL_WIDTH = 95
79 MODEL_HEIGHT = 95
80
81 # =====
82 # VISION CONFIDENCE / FALSE-MULTIPLE CONTROL
83 # =====
84 # User request
85 # - Ignores detections below 85% for normal confident decisions.
86 # - If two or more DIFFERENT vegetable labels are detected at 85% or higher,
87 # - show an asserted-vegetable error and black print/print.
88 # - If NO label reaches 85%, Fall back to the most frequent called prediction
89 # - From the model output, even if it is below 85%, so the device can still
90 # show the likely vegetable instead of always saying NO PRODUCT.
91 MIN_PRODUCT_CONFIDENCE = 0.85
92
93 # If True, weak secondary labels are ignored when there is only one strong label.
94 # Example: garlic:0.90 and marble_potato:0.60 becomes GARLIC only.
95 # However, the new asserted override below still wins when two labels are both
96 # -> MIN_PRODUCT_CONFIDENCE
97 VISION_BEST_LABEL_ONLY_MODE = True
98
99 # New requested behavior: high confidence different vegetables means asserted.
100 VISION_ASSERTED_WHEN_MULTIPLE_HIGH_CONF = True
101 VISION_ASSERTED_MESSAGE = "ERROR: ASSERTED DETECTED!"
102
103 # Fallback behavior when nothing reaches 85%.
104 # For FORD boxes, the code chooses the most frequent label.
105 # In broader: higher average confidence, then higher best confidence.
106 VISION_FALLBACK_TO MOST FREQUENT IF NO HIGH_CONF = True
107 VISION_FALLBACK_MIN_CONFIDENCE = 0.25
108 VISION_FALLBACK_MODEL_TEXT = "FREQUENT LOW CONF"
109
110 SMOOTHING_FRAMES = 5
111 INTER_INTERVAL = 0.25
112 CAMERA_SHARPEN_SEC = 1.8
113 CAPTURE_WIDTH = 960
114 CAPTURE_HEIGHT = 960
115
116 WINDOW_NAME = "Vegetable Detection - Weigh - Size 4:Print"
117 PREVIEW_WIDTH = 960
118 PREVIEW_HEIGHT = 960
119
120 # =====
121 # =====
122 # =====
123 # =====
124 # =====
125 # This uses the same square sensor-crop zoom logic as your separate IMX219
126 # zoom test code. At 1.0x, the camera uses a CENTERED SQUARE crop from
127 # the full IMX219 sensor, then scales it to 960 x 960.
128
129 # This is the important part for your dataset problem:
130 # - Preview and inference use the SAME square camera view.
131 # - 1.0x is loaded automatically at startup.
132 # - 1.0x here means the full centered square sensor crop, similar to your
133 # uploaded marble potato sample image.
134 # Higher values crop tighter from the center for sensor-level digital zoom.
135 CAMERA_ZOOM_DEFAULT = 1.00
136 CAMERA_ZOOM_MIN = 1.00
137 CAMERA_ZOOM_MAX = 4.00
138 CAMERA_ZOOM_STEP_FINE = 0.02
139 CAMERA_ZOOM_STEP_COARSE = 0.10
140 MIN_SCALE_CROP_SIZE = 64
141
142 EMPTY_LINE_LABELS = ("empty", "background", "none", "no object")
143
144 # =====
145 # =====
146 # =====
147 # =====
148 # =====
149 # Weight reading is now based on toAdWeight_3_lbn_0_ref_code/moving_average.py.
150 # Concept used:
151 # 1) median raw read
152 # 2) row-to-grain conversion from calibration
153 # 3) rolling trained average
154 # 4) stability window
155 # 5) display hold while moving
156 # 6) 3-decimal output with 0.1 g step
157 ROW_READS_PER_SAMPLE = 3
158 WEIGHT_READ_INTERVAL = 0.05
159
160 AVERAGE_WINDOW = 10
161 AVERAGE_TRIM_RATIO = 0.15
162
163 STABILITY_WINDOW = 8
164 STABLE_RANGE_G = 0.4
165 VERY_STABLE_RANGE_G = 0.2
166 STABLE_REQUIRED_COUNT = 4
167
168 DISPLAY_STEP_G = 0.1
169 DISPLAY_DECIMALS = 1
170
171 ZERO_DEADBAND_G = 0.3
172 NEGATIVE_CLAMP = False # False is required so soft tare can show negative weight when the tared load is removed.
173
174 DISPLAY_HOLD_WHEN_MOVING = True
175
176 DIO_CHANGE_UPDATE_G = 3.0
177
178 AUTOZERO_ENABLE = True
179 AUTOZERO_DEAD_G = 0.5
180 AUTOZERO_STABLE_RANGE_G = 0.25
181 AUTOZERO_CONSECUTIVE = 15
182 AUTOZERO_CORRECTION_G = 0.01
```



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```
start_thesis_device.sh  transaction_gui_crud.py  thesis_device_main.py  jekaca_pos_system_gui.py  thesis_device_main (6).py

381 AUTOZERO_CORRECTION_S = 0.01
382
383 # =====
384 # STARTUP AUTO-TARE SETTINGS
385 # =====
386 # Every time thesis_device_main.py starts, the current platform load becomes
387 # the display/not 0.0 g reading, so if the scale was tared
388 # This is a soft/NET tare, not a hard tare, so gross-load protection still
389 # uses the calibrated hard-zero baseline.
390 STARTUP_AUTO_TARE = True
391 STARTUP_TARE_READS = 20
392 STARTUP_TARE_DELAY_S = 0.03
393 STARTUP_TARE_MODE_TEXT = "STARTUP TARE"
394
395 # =====
396 # NEAR-ZERO AUTO-CORRECTION SETTINGS
397 # =====
398 # If the reading settles very close to zero after removing a load, this will
399 # automatically zero the microdisplay reading. This fixes cases where the scale
400 # stays at -1.0 g, 0.5 g, -0.7 g, etc. after the object is removed.
401 #
402 # It is protected by a stability check and consecutive count so it does not
403 # accidentally zero while a real object is being placed or removed.
404 NEAR_ZERO_AUTO_TARE_ENABLE = True
405 NEAR_ZERO_AUTO_TARE_MIN_S = -1.0
406 NEAR_ZERO_AUTO_TARE_MAX_S = 1.0
407 NEAR_ZERO_AUTO_TARE_STABLE_RANGE_S = 0.08
408 NEAR_ZERO_AUTO_TARE_CONSECUTIVE = 0
409 NEAR_ZERO_AUTO_TARE_COOLDOWN_S = 1.58
410 NEAR_ZERO_AUTO_TARE_MODE_TEXT = "AUTO ZERO"
411
412 WARNUP_READS = 5
413 WEIGHT_CONSOLE_UPDATE_INTERVAL = 0.20
414
415 # =====
416 # ZEMEC 10N load-cell safety. Tare does not remove actual physical load,
417 # so overload checks use gross load against the hard-zero baseline.
418 LOAD_CELL_MAX_CAPACITY_G = 8000.0
419 LOAD_CELL_WARNING_G = 7500.0
420 LOAD_CELL_OVERLOAD_G = 8000.0
421 BLINK_PRINT_WHEN_OVERLOAD = True
422
423 # =====
424 # LCD
425 # =====
426 LCD_I2C_ADDRESS = 0x27
427 LCD_I2C_PORT = 1
428 LCD_COLS = 16
429 LCD_ROWS = 2
430 LCD_REFRESH_INTERVAL = 0.20
431
432 # =====
433 # BUTTON / ACTUATOR / HITS960
434 # =====
435 BTN_PIN = 37
436 TARE_BTN_PIN = 27 # physical pin 33, active-HIGH
437 RPMW_PIN = 23
438 LPMW_PIN = 24
439 BUTTON_BOUNCE_S = 0.04
440 TARE_BUTTON_BOUNCE_S = 0.04
441 TARE_LOAD_PRESS_S = 1.50
442
443 # =====
444 # ADJUSTABLE ACTUATOR POSITION SETTINGS
445 # =====
446 # Because the linear actuator has NO position feedback sensor, all positions
447 # are controlled by time. You must calibrate these seconds on the real scalar.
448 #
449 # Desired default position when the OK/confirmation button is NOT pressed.
450 # User request: default should be maximum length / extend position.
451 # Allowed values: "extended" or "retracted"
452 DEFAULT_POSITION_BEFORE_PRESS = "extended"
453
454 # Position to move toward after the OK/confirmation button is pressed.
455 # If DEFAULT_POSITION_BEFORE_PRESS = "extended", this is normally "retracted".
456 # Allowed values: "extended" or "retracted"
457 POSITION_AFTER_PRESS = "retracted"
458
459 # Move time to force actuator toward default position
460 # Increase until it reaches your desired default/max extend position.
461 DEFAULT_POSITION_MOVE_TIME_S = 12.5
462
463 # Move time from default position toward position after press.
464 # This controls how far the actuator retracts/presses after button press.
465 POSITION_AFTER_PRESS_MOVE_TIME_S = 12.5
466
467 # Keep actuator stopped while scalar is pressed / in final pressed position.
468 PRESS_HOLD_TIME_S = 4
469
470 # Extra waiting time before returning to the default position.
471 RETURN_TO_DEFAULT_DELAY_S = 0.08
472
473 # =====
474 # Fuse between changing meter directions to protect the driver/actuator.
475 DIR_CHANGE_PAUSE_S = 0.18
476
477 # If True, the program moves actuator to DEFAULT_POSITION_BEFORE_PRESS at startup.
478 # Keep True if you want the actuator to automatically prepare itself before use.
479 MOVE_TO_DEFAULT_ON_STARTUP = True
480
481 # If True, it also moves toward the default position before every scaling cycle.
482 # Usually False is better after startup, because the actuator is already returned
483 # to default after each completed button cycle.
484 ENSURE_DEFAULT_BEFORE_EACH_CYCLE = False
485
486 # =====
487 # Safety
488 # =====
489 ALLOW_TRIGGER_WITHOUT_VALID_PRODUCT = False
490 MIN_WEIGHT_TO_ALLOW_PRINT_G = 1.0
491
492 # =====
493 # THERMAL PRINTER
494 # =====
495 PRINTER_ENABLED = True
496 PRINTER_DEVICE = "usb/lp0"
497 PRINT_FEED_LINES = 3
498 BARCODE_HEIGHT = 80
499 BARCODE_WIDTH = 5
500
501 # =====
502 # DATABASE SETTINGS
503 # =====
504 DEFAULT_VARIABLES = [
505     ("C", "0.0000", "100.0", "1")
506 ]
```



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start_thesis_device.sh x transaction_gui_crud.py x thesis_device_main.py x jekaca_pos_system_gui.py x thesis_device_main (6).py x
301 { "color": "ONLOW", 100, 0, 1},
302 { "color": "ONHI", 100, 0, 1},
303 { "color": "ONHI", 100, 0, 1},
304 }
305
306 # ===== LOCKS / STATE
307 # =====
308 state_lock = Lock()
309 printer_lock = Lock()
310 action_lock = Lock()
311 actuator_lock = Lock()
312
313
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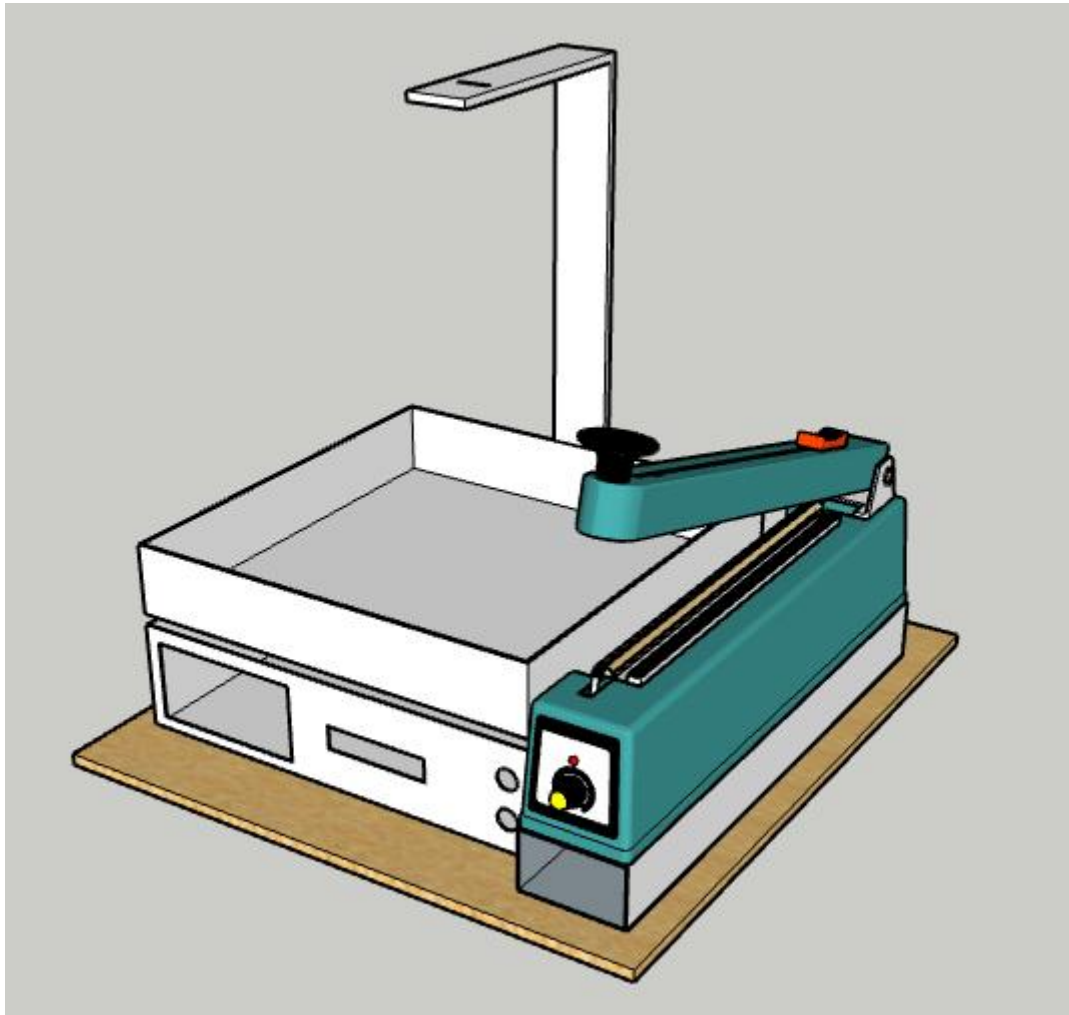
```
start_thesis_device.sh  transaction_gui_crud.py  thesis_device_main.py  jekaca_pos_system_gui.py  thesis_device_main (6).py

421 vegetable_name TEXT NOT NULL,
422 vegetable_display_name TEXT NOT NULL,
423 raw_detected_label TEXT NOT NULL,
424 confidence REAL NOT NULL,
425 weight_g REAL NOT NULL CHECK(weight_g >= 0),
426 price_per_kg REAL NOT NULL CHECK(price_per_kg >= 0),
427 total_price REAL NOT NULL CHECK(total_price >= 0),
428 transaction_time TEXT NOT NULL,
429 source_device TEXT NOT NULL DEFAULT 'raspberrypi_t_offline'
430 )
431
432 conn.execute()
433 conn.execute("CREATE INDEX IF NOT EXISTS idx_customer_transactions_barcode ON customer_transactions(barcode,san13)")
434
435 conn.execute("CREATE INDEX IF NOT EXISTS idx_customer_transactions_time ON customer_transactions(transaction_time)")
436
437 conn.commit()
438
439
440
441
442 def fetch_active_vegetables() -> dict[str, dict]:
443     """Return active vegetables keyed by normalized name."""
444     products: dict[str, dict] = {}
445     with get_conn(VDB_DB_PATH) as conn:
446         rows = conn.execute("""
447             SELECT name, display_name, price_per_kg
448             FROM vegetables
449             WHERE is_active = 1
450             ORDER BY name
451         """)
452         for row in rows:
453             products[row["name"].strip().lower()] = {
454                 "display_name": row["display_name"],
455                 "price_per_kg": float(row["price_per_kg"]),
456             }
457         return products
458
459
460 def create_san13_from_id(txn_id: int) -> str:
461     """
462     Build a numeric-only EAN13 barcode from the transaction id.
463     Base 12 digits = transaction id zero-padded to 12 digits.
464     Digit 13 = EAN13 check digit.
465     """
466     base12 = f"{txn_id:012}"
467     return base12 + san13_check_digit(base12)
468
469
470 def insert_transaction(snapshot: SystemState) -> tuple[int, str]:
471     """Insert a transaction and generate a unique EAN13 barcode from the DB id."""
472     transaction_time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
473     with get_conn(TXN_DB_PATH) as conn:
474         cursor = conn.cursor()
475         cursor.execute("""
476             INSERT INTO customer_transactions (
477                 vegetable_name, vegetable_display_name, raw_detected_label, confidence, weight_g, price_per_kg, total_price, transaction_time, source_device
478             ) VALUES (
479                 ?, ?, ?, ?, ?, ?, ?, ?, ?
480             )
481         """)
482         txn_id = cursor.lastrowid
483         san13 = create_san13_from_id(txn_id)
484         return (txn_id, san13)
```



APPENDIX D

PROPOSED DESIGN





APPENDIX E

LETTERS FOR VALIDATION



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April 22, 2026

Ms. Charmaine Agustin - Sabio, MaEd
Carlos Hilado Memorial State University
Barangay Alijis, Bacolod City, 6100

Dear Ma'am:

The undersigned is a fourth year Bachelor of Science in Computer Engineering (BSCpE) of Carlos Hilado Memorial State University undertaking a research entitled "Automated Produce Scanner with Real-time Pricing and Packaging".

With your expertise, I am humbly asking your permission to validate the attached self-made questionnaire, for the study using the attached rating tool

Your presence will be highly appreciated.

Thank you very much. God Bless!

Respectfully yours,


John Carlo T. Gonzales, et al.

Noted by:


Alfonso Jorie Benedicto III
Research Adviser


Engr. Josephine B. Salillas, PhD
Practical Design 2 Professor



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April 22, 2026

Ms. Christine Bustamante
Carlos Hilado Memorial State University
Barangay Alijis, Bacolod City, 6100

Dear Ma'am:

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Practical Design 2 Professor



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College of Engineering

April 22, 2026

Ms. Juliet Samillano
Carlos Hilado Memorial State University
Barangay Alijis, Bacolod City, 6100

Dear Ma'am:

The undersigned is a fourth year Bachelor of Science in Computer Engineering (BSCpE) of Carlos Hilado Memorial State University undertaking a research entitled "Automated Produce Scanner with Real-time Pricing and Packaging".

With your expertise, I am humbly asking your permission to validate the attached self-made questionnaire, for the study using the attached rating tool

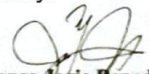
Your presence will be highly appreciated.

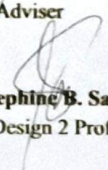
Thank you very much. God Bless!

Respectfully yours,


John Carlo T. Gonzales, et al.

Noted by:


Alfonso Jorie Benedicto III
Research Adviser


Engr. Josephine B. Salillas, PhD
Practical Design 2 Professor



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April 22, 2026

Dr. Richard Pabelona Jr.
College of Engineering
Carlos Hilado Memorial State University
Barangay Alijis, Bacolod City, 6100

Dear Sir:

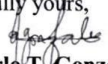
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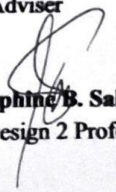
Thank you very much. God Bless!

Respectfully yours,


John Carlo T. Gonzales, et al.

Noted by:


Alfonso Jorie Benedicto III
Research Adviser


Engr. Josephine B. Salillas, PhD
Practical Design 2 Professor



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April 22, 2026

Ms. Evalyn Pedrosa
Carlos Hilado Memorial State University
Barangay Alijis, Bacolod City, 6100

Dear Ma'am:

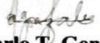
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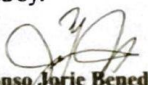
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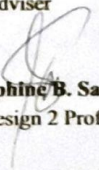
Thank you very much. God Bless!

Respectfully yours,


John Carlo T. Gonzales, et al.

Noted by:


Alfonso Jorie Benedicto III
Research Adviser


Engr. Josephine B. Salillas, PhD
Practical Design 2 Professor



APPENDIX F

RESEARCH EVALUATION INSTRUMENT



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Dear respondents:

We are conducting a study to design, develop, and evaluate Automated Produce Scanner with Real-time Pricing and Packaging.

The purpose of this survey is to gather your insights and evaluation regarding the system's performance, functionality, usability, and overall effectiveness. Your responses will help assess and improve the proposed system.

This questionnaire will take approximately 5–10 minutes to complete. There are no right or wrong answers; we encourage you to answer honestly based on your experience and understanding.

All information collected will be treated with strict confidentiality and will be used solely for academic and research purposes.

By answering this questionnaire, you voluntarily agree to participate in this study.

Thank you for your time and valuable participation!

Part I: Respondent Profile

Age: ____ Gender: ____ Field/Specialization : ____

Part II: Questionnaire Evaluation

Instruction: Please indicate your degree of agreement or disagreement on the statements provided below by encircling the number which corresponds to your best to your judgment.

1 – Strongly Disagree 2 – Disagree 3 – Undecided 4 – Agree 5 – Strongly Agree

A. Clarity of Questions	1	2	3	4	5
1. The questions are clear and easy to understand.					
2. The terms used are appropriate for Computer Engineering students and vendors.					
3. The instructions are easy to follow.					
4. The questions are not confusing or ambiguous.					
B. Relevance					
5. The questions are relevant to computer engineering devices and real-world applications					
6. The items reflect actual concepts used in designing and developing computer engineering systems and applications.					
7. Overall, the questionnaire is appropriate and relevant for evaluating computer engineering devices and applications.					
C. Consistency					



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8. The questions consistently measure the same concept.					
9. Similar questions provide consistent responses.					
10. There are no contradictory items.					
D. Structure					
11. The questionnaire is well-organized.					
12. The sequence of questions is logical.					
13. The length is appropriate.					
E. Overall					
14. I was able to answer without difficulty.					
15. The questionnaire is suitable for research.					

Part III: Comments

Please write your comments/suggestions:

Evaluated by: (Please sign below, you may opt not to include your name for data privacy)

Signature/Date of Evaluation

Please contact us for further information: carlogonzales1112345@gmail.com

Thank you for your interest and time!



APPENDIX G

STATISTICAL TOOLS

Mean:

$$\bar{x} = \frac{\sum x}{n}$$

Standard Deviation:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Cronbach Alpha:

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum_{i=1}^k \sigma_i^2}{\sigma_X^2} \right)$$



APPENDIX H

USER'S

MANUAL

AUTOMATED PRODUCE SCANNER WITH REAL-TIME PRICING AND PACKAGING

User Manual



BRILLAS | GONZALES | JAMELANO | LACSON | MAGAN

MAY 2026 VERSION 1.0

1

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Table of Contents	3
Features	4
Safety Instructions	5
Description of Parts	7
Operation Procedure	8
Technical Specification	9

3

PREFACE

About this Manual

This user manual has been created to help you understand and safely operate the device. For the best results, please read all sections carefully before using the scanner, and keep this manual in a safe place for future reference.

About the Device

The Automated Produce Scanner with Real-Time Pricing and Packaging is a self-service machine that identifies vegetables, weighs them, calculates the price, seals them in a bag, and prints a barcode for checkout using a camera, machine learning, and a built-in scale and sealer.

How This Manual is Organised

- Features – key capabilities of the device.
- Safety Instructions – important warnings and safe-handling rules.
- Description of Parts – purpose of each component.
- Operation Procedure – step-by-step guide from start-up to transaction.
- Technical Specifications – detailed technical data.

Target Audience

This manual is written for end-users (shoppers), store administrators, and technical support personnel. Administrators should pay special attention to the "Admin Price Update" section.

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FEATURES

The Device is designed to simplify the purchase of loose vegetables. Its main features are:

- Vegetable classification – identifies red onion, garlic, and marble potato.
- Assorted detection – detects mixed vegetables and shows an error message.
- Automatic weighing – measures weight from 100 g to 3 kg (± 5 g tolerance).
- Real-time pricing – calculates total cost based on stored unit prices.
- Updatable prices – allows admin to modify prices instantly.
- Automated sealing – seals the bag after purchase confirmation.
- Barcode printing – generates and prints a unique EAN-13 barcode.
- User interface – displays details and uses buttons for tare and confirmation.
- Stand-alone operation – runs locally on a Raspberry Pi without internet.

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SAFETY INSTRUCTIONS

A. For General Public (Shoppers)

- Read manual before use.
- Place only one vegetable type.
- Maximum weight: 3 kg.
- Do not lean on platform.
- Sealer gets hot – do not touch.
- Wait 10 s before removing bag.
- No liquids near device.
- Use only supplied LDPE bags.
- Keep fingers away from moving parts.
- Report errors to staff – do not repair.

B. For Administrators (Maintenance Personnel)

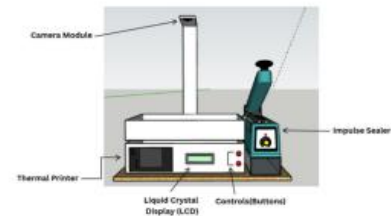
- Disconnect power before service.
- Only trained personnel open housing.
- Let sealer cool 5 min before touching.
- Update prices via authorised interface only.
- Do not edit database directly.
- Recalibrate load cell after repairs.
- Keep ventilation slots unblocked.
- Unplug and wait 1 min before internal work.
- Do not power off during firmware updates.
- Inspect sealer and printer monthly.
- Use only approved spare parts.

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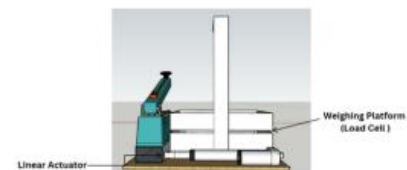
DESCRIPTION OF PARTS

The diagram below (Figure 1) shows the physical arrangement; the table that follows describes the role of each part.

FRONT VIEW



BACK VIEW



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DESCRIPTION OF PARTS



Main Unit (Raspberry Pi 4 Model B)
acts as the central controller, it handles weight reading, price calculation components. It runs on a pre-installed direct user setup is needed.



Camera Module
mounted above the platform to capture images for identification. It is pre-calibrated and should be kept clean.



Weighing Platform (Load Cell)
measures the weight of items and supports 100 g to 3 kg with ± 5 g.



Liquid Crystal Display (LCD)
provides real-time information such as confidence level, weight, total price messages.



Impulse Sealer and Linear Actuator
automatically seals the plastic bag by pressing it against a heated element.



Thermal Printer
prints a barcode receipt (EAN-13) for easy checkout scanning, using a barcode.



Controls (Buttons)
includes a tare button to reset the scale to complete the transaction.



Power Supply
integrates 5 V for the Raspberry Pi actuator into a single, easy-to-use power source.

OPERATION PROCEDURE

Normal operation is a simple sequence of actions. Follow these steps to weigh, price, seal and obtain a barcode for a vegetable item.

STEP 1

Power On and Startup – connect the device to power and turn it on. The LCD will display a welcome message while the system initializes. Wait about 15 seconds for full startup.

STEP 2

Place the Vegetable – put one type of vegetable on the platform. The system will scan and identify it. If mixed items are detected, an error appears and you must remove extras.

STEP 3

Check LCD Results – the screen shows the vegetable name, confidence level, weight, and total price based on stored data.

STEP 4

Confirm the Transaction – press the confirm button if the details are correct. If not, remove and place the item again to recalculate.

STEP 5

Sealing and Printing – the device automatically seals the bag and prints a barcode label after confirmation. Remove the sealed item and keep the barcode for checkout.

STEP 6

Admin Price Update – authorized users can update vegetable prices through the system interface, and changes apply immediately.

STEP 7

Tare Function – press the tare button to reset the scale to zero, useful when using containers or bags before weighing.

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TECHNICAL SPECIFICATION

Category	Specification
Model	Automated Produce Scanner v1.0
Microcontroller	Raspberry Pi 4 Model B (4 GB RAM)
Processor	Broadcom BCM2711, quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.8 GHz
Camera	Raspberry Pi Camera Module V2 (IMX219) - 8 megapixels, fixed focus
Recognition architecture	FOMO (Faster Objects, More Objects) on Edge Impulse
Supported vegetables	Red onion, garlic, marble potato (3 classes)
Weighing sensor	Load cell (10 kg capacity) + HX711 24-bit ADC
Weighing range	100 g to 3 kg

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TECHNICAL SPECIFICATION

Weighing accuracy	±5 g (tolerance)
Price calculation	Real-time, based on local unit price (PHP/kg)
Display	16 × 2 character LCD, I2C interface, blue backlight
Sealing system	Impulse heat sealer, actuated by 12 V DC linear actuator (150 mm stroke)
Sealable bag material	Low-density polyethylene (LDPE) heat-sealable bags
Printer type	Direct-thermal receipt printer
Barcode symbology	EAN-13 (13 digits)
Paper width	58 mm or 80 mm (depending on configuration)
User controls	Two push-buttons: Tare, Confirm

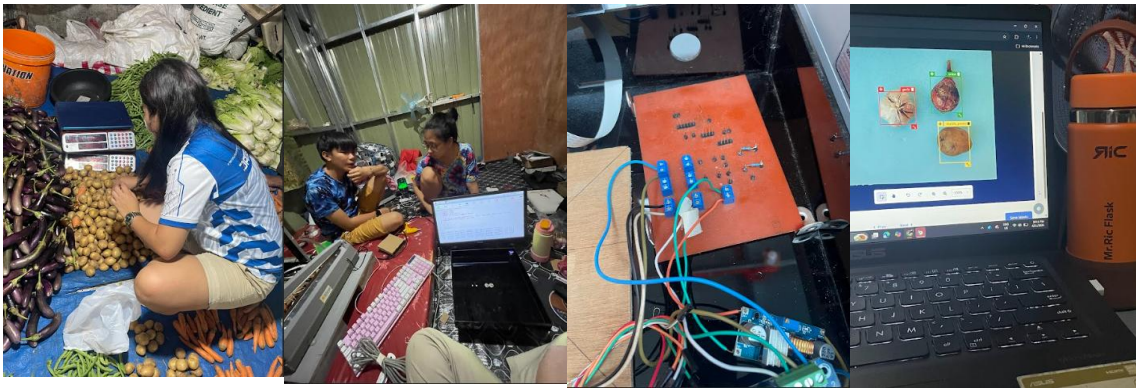
10

TECHNICAL SPECIFICATION

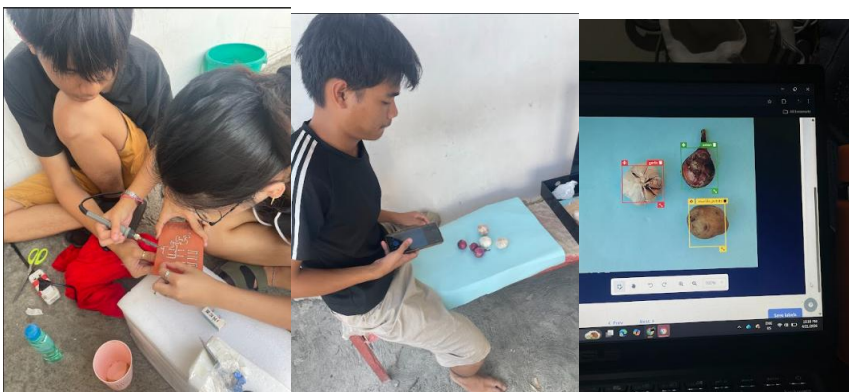
Database	Local SQLite database (off-line)
Power supply	Primary: 5 V / 3 A (for Raspberry Pi); Secondary: 12 V DC (for sealer and actuator)
Power consumption	Maximum 50 W during sealing
Operating temperature	10 °C to 35 °C
Storage temperature	-10 °C to 50 °C
Relative humidity	20 % to 80 % (non-condensing)
Dimensions (approx.)	400 mm (W) × 350 mm (D) × 400 mm (H)
Weight (empty)	Approximately 5 kg



APPENDIX I

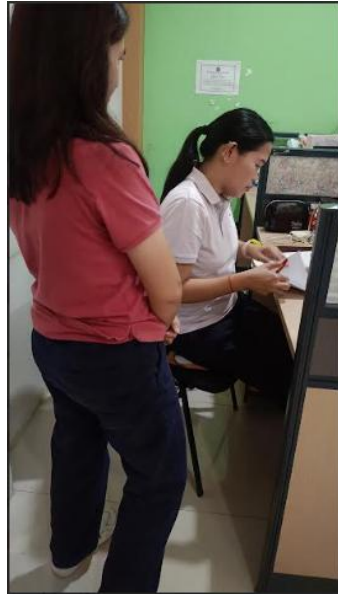


DOCUMENTATION

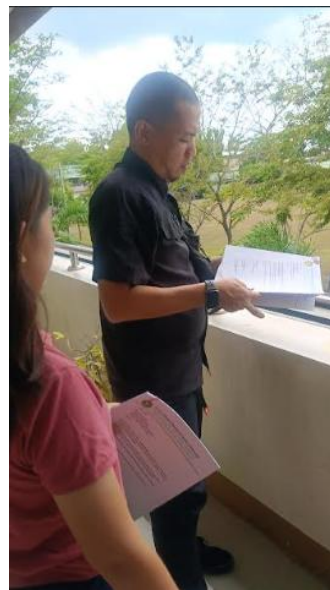




APPENDIX J



EVALUATION FOR VALIDATION AND RELIABILITY





APPENDIX K

CURRICULUM VITAE (*Researcher's Profile*)

MARIA JENI LOUISE BRILLAS

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Email: brillasmariajenilouise@gmail.com

Address: 1st Street, Purok Lirio, Barangay Taculing,
Bacolod City 6100, Negros Occidental



OBJECTIVE:

I am a motivated and an enthusiastic Computer Engineering student seeking an internship opportunity in a dynamic environment where I could practice my knowledge and develop my personality as a career person while utilizing my skills.



TECHNICAL SKILLS:

- Microsoft Office (Excel, Word, PowerPoint)
- Can interpret Circuits and Design
- Hands-on experience in basic Kali Linux navigation and operations
- Experienced in practical breadboarding and circuitry
- Basic programming

SOFT SKILLS:

- Communication Skills
 - Good listening skills
 - Time Management Skills
 - Organizational Skills
 - Ability to work independently or as part of a team
 - Problem-Solving and Critical Thinking
 - Decision-Making
-

EDUCATION:

Tertiary: Bachelor of Science in Computer Engineering (2022 - Present)
Carlos Hilado Memorial State University – Alijis, Bacolod City

Secondary: Bacolod City National High School (2016 – 2022)
Taculing, Bacolod City

Elementary: Crispino V. Ramos Elementary School (2011 – 2016)
Taculing, Bacolod City
Marikina Catholic School (2009 - 2010)
Marikina City

CERTIFICATION:

MATLAB: Data Analysis

MathWorks| Completed Nov 2024

Tables, Clean and Prepared Data for Analysis, Common Data Analysis Techniques Find and Extract Subsets of Data, and Calculations on Grouped Data



One Negros Ads

Robotics Workshop Instructor (Student Intern)

PROJECTS & EXPERIENCES:

High School Research Paper

Division Scilympics Champion, Best Paper (Physical Science Team Category)

”Synergized *Caulerpa lentillefera* Lato and *Eucheuma cottonii* Gusó seaweeds dried as alternative soil conditioners and its Fertilizing Potential”

Google Devfest Bacolod

Google Devfest Bacolod I/O Extended 2024 – Attendee

Google Devfest Bacolod Bacolod 2024 – Attendee

Google Devfest Bacolod 2025 – Attendee

John Carlo Gonzales

Phone : +639667355370

Email : gonzales.johncarlo.t@gmail.com

Address : San Pedro Street, Barangay Rumirang,
Isabela, Negros Occidental



OBJECTIVE:

I am a dedicated Computer Engineering student seeking an internship opportunity to further develop my technical abilities while gaining real-world experience in a professional environment.

TECHNICAL SKILLS:



Basic Programming (C, C++, Python)
Circuit analysis and design
Breadboarding and hardware troubleshooting
Arduino / basic embedded systems
Verilog / VHDL (basic)
Windows & Linux
Microsoft Word, Excel, PowerPoint; simulation and debugging tools

SOFT SKILLS:

Adaptability and Willingness to Learn
Time Management and Task Prioritization
Attention to Detail
Analytical and Problem-Solving Ability
Strong Communication Skills
Team Collaboration and Coordination

EDUCATION:

Tertiary: Bachelor of Science in Computer Engineering (2022 - Present)
Carlos Hilado Memorial State University – Alijis, Bacolod City, Negros Occidental

Secondary: Isabela National High School (2016 – 2022)
Rizal Extension Street, Brgy. 7, Isabela, Negros Occidental

Elementary: Rumirang Elementary School (2010 – 2016)
Barangay Rumirang, Isabela, Negros Occidental

CERTIFICATION & TRAINING:

One Negros Ads

Robotics Workshop Instructor (Student Intern)

Hack The Box Academy – Cybersecurity Fundamentals

Hands-on training in basic Linux, networking, and security concepts

Cisco Networking Basics – Cisco Academy | 2025

Network fundamentals, IP addressing, router configuration, troubleshooting



JEAN ROSE S. JAMELANO

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Address: ECC Villas, Brgy. Alijis,
Bacolod City, 6100, Negros Occidental



OBJECTIVE:



Dedicated and motivated Computer Engineering student seeking an opportunity to apply theoretical knowledge in practical settings, contribute to organizational goals, and develop professional skills that connect to engineering fields.

TECHNICAL SKILLS:

- Have knowledge in Network Communication, IP Addressing, Cisco Networking Basics
- Basic Programming
- Microsoft Office (Word, PowerPoint)
- Hands-on experience in Linux (Ubuntu, Kali)

SOFT SKILLS:

- Communication Skills
- Good listening skills
- Time Management Skills
- Decision Making
- Ability to work independently or as part of a team

EDUCATION:

Tertiary: Bachelor of Science in Computer Engineering (2023 - Present)

Carlos Hilado Memorial State University – Alijis, Bacolod City

: Bachelor of Science in Computer Engineering (2021 - 2022)

Systems Plus College Foundation, Balibago, Angeles City, Pampanga

Secondary: Holy Trinity School:Center of Catholic Education Inc. (2019 - 2021)

Manibaug Paralaya, Porac Pampanga

: Victorias National High School (2015 - 2019)

Yap Quiña Street, Victorias City, Negros Occidental

Elementary: Nanca Elementary School (2010 - 2015)

Brgy. Nanca E.B. Magalona , Negros Occidental

CERTIFICATION:



MATLAB: Data Analysis

MathWorks| Completed Nov 2024

Tables, Clean and Prepared Data for Analysis, Common Data Analysis Techniques Find and Extract Subsets of Data, and Calculations on Grouped Data

Google Devfest Bacolod

Google DevFest 2024

Attendee - **Google Developer Group** |December 2024

GDG Bacolod – “Google I/O Extended Bacolod 2025”

Attendee - **Google Developer Group Bacolod** | November 2025

One Negros Ads

Robotics Workshop Instructor (Student Intern)

JESTON ROEM C. LACSON

Phone: +639667767792

Email: jestionroem@gmail.com

Address: Purok Sabes, Brgy. Villamonte,
Bacolod City, 6100, Negros Occidental





OBJECTIVE:

Dedicated to becoming a skilled Computer Engineering, I aim to apply my academic knowledge, enhance my technical abilities, and contribute positively to organizational success.

TECHNICAL SKILLS:

- PC assembly, troubleshooting repair
- Operating system installation / reinstallation (Windows & Linux)
- Microsoft Office (Word, PowerPoint, Excel)
- Hands-on experience in Linux (Ubuntu, Kali)
- Soldering circuit components
- Router set up Ethernet RJ45 & UTP cable crimping and splicing

SOFT SKILLS:

- Communication Skills
- Teamwork and Collaboration
- Quick to learn and adaptable to new situations

EDUCATION:

Bachelor of Science in Computer Engineering

- September 2021-January 2023 Carlos Hilado Memorial State University
Science, Technology, Engineering and Mathematics (STEM)
- August 2019 - May 2021

Bacolod City National Highschool

- Graduated With Honors
- Student Service Awardee

Luis Hervias National Highschool

- Graduated With Honors

Credentials:

- Common Data Analysis Techniques – MATLAB Academy, 1 hr
- Calculations on Grouped Data – MATLAB Academy, 1 hr



- Find and Extract Subsets of Data – MATLAB Academy, 1 hr
- Clean and Prepare Data for Analysis – MATLAB Academy, 1.5 hrs
- Tables – MATLAB Academy, 2 hrs
- ROBOTICS WORKSHOP INSTRUCTOR - Student Intern (One Negros Ads)

KANE MAXINE AESHA M. MAGAN

Phone: +63946027395

Email: kanemaxineaesham@gmail.com

Address: Greenplains Subd., Alijis Road
Bacolod City, Negros Occidental



OBJECTIVE:

A motivated and detail-oriented Computer Engineering student seeking an opportunity where I can apply my skills in robotics, circuit design, breadboarding, and hardware–software integration. Eager to gain hands-on experience and contribute to real-world engineering projects, including embedded systems, networking solutions, and cybersecurity-related initiatives.



TECHNICAL SKILLS:

- Hardware, Circuits & Robotics: Circuit interpretation and basic design, intermediate breadboarding, practical prototyping, introductory robotics systems
- Networking & Security: Cisco Networking Basics, IPv4/IPv6 addressing, network troubleshooting, cloud security fundamentals
- Operating Systems & Virtualization: Windows, Linux (Ubuntu, Kali Linux navigation), virtual machine setup and management
- Programming & Tools: MATLAB (data analysis), Python (basic scripting), Microsoft Office Suite

SOFT SKILLS:

- Analytical Thinking & Problem-Solving
 - Detail-Oriented & Organized
 - Collaborative Team Player
 - Adaptable to New Tools & Environments
 - Strong Written & Verbal Communication
-

EDUCATION:

Tertiary : Bachelor of Science in Computer Engineering (2022 - Present)
Carlos Hilado Memorial State University – Alijis, Bacolod City

Secondary : Doña Hortencia Salas Benedicto National High School (2016 – 2022)
Burgos Street, La Carlota City, Negros Occidental

Elementary : La Carlota South Elementary School - 1 (2011 – 2016) Gurrea Street, La Carlota City, Negros Occidental

PROFESSIONAL DEVELOPMENT & AI WORKSHOPS

Google DevFest 2024

Attendee - Google Developer Group | December 2024

- Attended keynote sessions and technical workshops on modern web and mobile development, AI, and cloud integration.
- Gained insight into Google developer ecosystems and scalable tools such as Firebase and GenAI.
- Explored emerging blockchain technologies through a focused session on decentralized solutions and realworld applications.



- Networked with local developers and professionals to expand industry awareness and connections.

Google DevFest 2025

Attendee - Google Developer Group | November 2025

- Explored practical applications of Realtime RAG, Gemini, and the Multimodal Live API through live demos and guided sessions.
- Learned how AI is being integrated into real-world solutions across startups, education, and creative industries.
- Collaborated with fellow developers to ideate and prototype AI-driven solutions during breakout activities.

CERTIFICATIONS:

MATLAB: Data Analysis – MathWorks | Nov 2024

- Data cleaning, visualization, group calculations, filtering
- Cisco Networking Basics – Cisco Academy | 2025 Network fundamentals, IP addressing, router configuration, troubleshooting
- Robotics Instructor - Student Intern (One Negros Ads)